生物信息学简报，2025年，26(5)，bbaf498
https://doi.org/10.1093/bib/bbaf498
问题解决协议

# TCM-navigator，一种基于深度学习的工作流程，用于生成和评估类似传统中药的化合物以进行药物开发

陈飞鹰 [1,2,‡], 林俊宇 Victor [2,‡], 李明宇 [1],范浩 [2,3,*]

[1]上海交通大学医学院药物化学与生物信息学中心，中国上海 200025[2]生物信息学研究所 (BII)，新加坡科学、技术与研究局 (A*STAR)，30 Biopolis Street, #07-01 Matrix Building，新加坡 138671， 新加坡共和国[3]杜克-国大医学院，8 College Rd，新加坡 169857，新加坡共和国

*通信作者。新加坡科学技术研究局（A*STAR）生物信息学研究所（BII），新加坡碧波路30号矩阵大厦07-01室，邮编138671，新加坡共和国。电子邮件：hfan2006@gmail.com
‡陈飞英和林俊宇对这项工作贡献相同。

## 摘要

传统中医药（TCM）长期以来被视为现代药物发现的重要资源。然而，记录实体和信息的有限可用性、草药–成分–靶点–疾病网络的复杂性和稀疏性，以及数据表示的不一致性，阻碍了高通量筛选方法的有效性。尽管通过人工实验筛选已经发现了一些具有治疗价值的中药化合物，但这些方法耗时且需要大量人力资源。为了解决这些挑战，我们开发了一个以数据驱动和深度学习为基础的工作流程——TCM-navigator，该流程能够进行中药类分子的计算机生成、质量控制和基于物理的评估。生成过程由TCM-Generator完成，这是一种基于迁移学习和长短期记忆（LSTM）的化学语言模型，可生成标准化、分层结构化且适合高通量处理的中药类分子数据集。

关键词：中医；TCM导航器；TCM生成器；TCM识别器；深度学习；化学语言模型

## 介绍

传统中医（TCM）经过数千年的广泛实践而被探索、发现和验证，涵盖了预防、治疗和健康调理等过程 [1]。统计数据显示，大约三分之一的现代药物来源于天然草药[2]。加上天然化合物普遍低毒性，中医药具有巨大的潜力，并已成为现代药物开发的重要资源。值得注意的是，某些中药化合物已经在成分和分子水平上进一步探索和验证，从而成功应用于临床治疗。最著名的例子是屠呦呦诺贝尔奖发现的青蒿素，它已广泛用于治疗疟疾[3]。近年来，中医药的现代化受到了显著关注。为了规范和统一现有的中医药知识方面，研究人员已经建立了高质量的中医相关数据库，如 TCMBank [2]、ETCM [4]，和 TCM-ID [5]。中医数据库的出现使得数据驱动的方法得以加速中医相关药物的发现，弥合了古代实践与现代药理学之间的差距[6]。尽管这些知识库加深了研究人员的理解并激发了进一步探索的兴趣，但中医在当代药物开发中仍然是一个未充分利用的资源。这种不足利用源于若干内在挑战。中医的运作涵盖多个层面——从宏观的草药到微观的分子——其天然来源导致了特别复杂的药材–成分–靶点–疾病网络。这种复杂性导致成分供应有限、分子组成复杂且作用机制大多尚未解决。这些特征阻碍了中医知识库的系统性和高通量利用。此外，

收到日期：2025年5月30日。修订日期：2025年8月17日。接受日期：2025年9月3日 

*Briefings in Bioinformatics*, 2025, **26(5)**, bbaf498
https://doi.org/10.1093/bib/bbaf498
Problem Solving Protocol

# TCM-navigator, a deep learning-based workflow for generation and evaluation of traditional Chinese medicine-like compounds for drug development

Feiying Chen [1,2,‡], Victor Jun Yu Lim [2,‡], Mingyu Li [1], Hao Fan [2,3,*]

[1]Medicinal Chemistry and Bioinformatics Center, Shanghai Jiao Tong University School of Medicine, Shanghai 200025, China
[2]Bioinformatics Institute (BII), Agency for Science, Technology and Research (A*STAR), 30 Biopolis Street, #07-01 Matrix Building, Singapore 138671, Republic of Singapore
[3]Duke-NUS Medical School, 8 College Rd, Singapore 169857, Republic of Singapore

*Corresponding author. Bioinformatics Institute (BII), Agency for Science, Technology and Research (A*STAR), 30 Biopolis Street, #07-01 Matrix Building, Singapore 138671, Republic of Singapore. E-mail: hfan2006@gmail.com
‡Feiying Chen and Victor Jun Yu Lim contributed equally to this work.

## Abstract

Traditional Chinese Medicine (TCM) has long been regarded as a valuable resource for modern drug discovery. However, the limited availability of recorded entities and information, the complexity and sparsity of the herb–ingredient–target–disease network, and inconsistencies in data representation hinder the effectiveness of high-throughput screening approaches. While some therapeutically valuable compounds from TCM have been discovered through manual experimental screening, such methods are time-consuming and require substantial human resources. To address these challenges, we developed a data-driven and deep learning–based workflow, TCM-navigator, which enables the *in-silico* generation, quality control, and physics-based evaluation of TCM-like molecules. The generation is done by TCM-Generator, a transfer learning- and Long Short-Term Memory (LSTM)-based chemical language model that generates standardized, hierarchically structured, and high-throughput–friendly datasets of TCM-like molecules. In this study, we generated a target-nonspecific dataset comprising 3.7 million TCM-like molecules, expanding the number of entities in existing TCM datasets by more than 100-fold. The workflow also enables flexible, goal-driven molecule generation customized for specific targets, yielding three target-specific datasets and multiple high-potential target-ligand pairs. The quality control is done by TCM-Identifier, the first quantitative model specifically designed to capture unique characteristics of TCM, using an AttentiveFP framework with message passing neural networks. TCM-Identifier is expected to serve as an essential evaluation and guidance tool for TCM-related drug development. Our workflow bridges cutting-edge data science—including deep learning—with biomedical research to tackle longstanding challenges in target identification and molecular design. Its adaptable framework is also transferable to interdisciplinary innovation beyond drug development.

**Keywords:** traditional Chinese medicine; TCM-navigator; TCM-generator; TCM-identifier; deep learning; chemical language model

## Introduction

Traditional Chinese Medicine (TCM) has been explored, discovered, and validated through extensive practice for thousands of years, encompassing processes such as prevention, treatment, and health regulation [1]. Statistics indicate that approximately one-third of modern medicines are derived from natural herbs [2]. Coupled with the generally low toxicity of natural compounds, TCM holds immense potential and has become an important resource for modern drug discovery. Notably, certain TCM compounds have undergone further exploration and validation at the ingredient and molecular levels, leading to their successful application in clinical treatments. The most well-known example is artemisinin, discovered by Nobel Laureate Tu Youyou, which has been widely adopted as an effective treatment for malaria [3]. In recent years, the modernization of TCM has gathered significant attention. To standardize and unify existing TCM knowledge, researchers have established high-quality TCM-related databases, such as TCMBank [2], ETCM [4], and TCM-ID [5]. The emergence of TCM databases has enabled data-driven approaches that accelerate TCM-related drug discovery, bridging the gap between ancient practices and modern pharmacology [6]. Although these knowledge bases have deepened researchers' understanding and inspired further exploration, TCM remains an underexploited resource in contemporary drug development. This underuse stems from several intrinsic challenges. TCM operates across multiple scales—from macroscopic herbs to microscopic molecules—and its natural origins give rise to a particularly complex herb–ingredient–target–disease network. This complexity results in limited ingredient availability, intricate molecular compositions, and largely unresolved mechanisms of action. These characteristics hinder the systematic and high-throughput utilization of TCM knowledge bases. Moreover,

**Received:** May 30, 2025. **Revised:** August 17, 2025. **Accepted:** September 3, 2025

人工实验验证耗时且劳动密集，进一步减缓了中医相关药物研发的进展，并限制了其临床转化潜力。

为了解决上述问题，我们假设经过大量化学库预训练并通过中药复合物数据库微调的化学语言模型，能够生成具有类似原始中药分子（中药样）生物、化学或药理特性的全新化学物质。随后，我们开发了TCM-navigator，这是第一个基于深度学习的端到端工作流程，连接传统中药知识与现代药物发现流程。TCM-navigator 能够实现中药样化合物的生成、质量控制和基于物理的评估。它包含三个核心模块：（i）TCM-generator，通过迁移学习和LSTM架构利用中药知识库中的现有知识，扩展并完善中药数据库的原始化学空间，从而灵活适应特定靶点的中药样化合物生成；（ii）TCM-identifier，第一个定量中药评分模型，利用Attentive FP框架

为了展示我们类似中药化合物的应用，我们准备了一个案例研究，该研究包括三个重要的药物靶点，包括用于癌症的表皮生长因子受体（EGFR）、用于病毒感染的SARS-CoV-2主蛋白酶（Mpro），以及用于糖尿病和代谢性疾病的Sirtuin 1（SIRT1）。这三个靶点在治疗各种疾病方面具有重要的临床应用，并且已被中药分子 [7–9]靶向。在我们的案例研究中，来自类似中药数据集的分子显示出比已知与这些靶点相互作用的中药分子更好的预测结合潜力，这表明它们是有前景的配体候选物。

除了加速中药相关药物的开发，更重要的是，我们的工作流程解决了数据驱动生物学中的一个关键挑战：标准化和扩展有限的数据集。尽管高通量技术和深度学习驱动的流程日益普及，在药物化学和合成生物学等领域，小到中等规模的数据集仍常常存在标准化不足、可扩展性有限以及数据不完整的问题，使得人工处理效率低下且容易出错。我们的工作流程通过支持计算机模拟分子生成、质量控制和基于物理的评估来应对这一问题，将稀疏数据转化为丰富、标准化的实验探索资源。这种综合方法统一了多样的生物医学领域，实现了可扩展、高效和简化的研究。

## 结果与讨论

我们提出了一个端到端的工作流程——TCM导航器（TCM-navigator）——用于类似中药复合物的生成和评估，该流程整合了多个数据驱动和基于深度学习的组件。我们的工作流程组织为三个核心设计模块：两个创新组件，包括TCM生成器（TCM-generator）（图1A）和TCM识别器（TCM-identifier）（图1B），以及基于物理的评估（图1C）。TCM生成器是一个大规模分子生成模块，采用由三个阶段组成的分阶段学习策略

阶段：预训练、微调和状态调优。这个分阶段的方法使TCM生成器能够逐步编码不同化合物库的化学特征，实现各阶段的高效特征提取[10]。TCM识别器是基于AttentiveFP的分子鉴别器，开发用于评估生成化合物的中药相似性[11]。作为首个用于中药类性质评估的深度学习模型，TCM识别器能够系统地捕获分子的整体信息，同时保持高可解释性。在基于物理的评估模块中，我们采用两个关键标准：（i）静态结合亲和力——采用分子对接预测结合构象，并评估药物分子与靶蛋白的结构和化学互补性；（ii）动态结合稳定性——通过分子动力学模拟评估入选中药类化合物在结合口袋内的稳定性。我们的工作流程建立了一种创新的端到

## TCM-生成器：用于生成中药类分子的化学语言模型

中药样分子的生成。为了使化学语言模型能够全面学习SMILES序列的语法和特征分布，我们在ZINC15筛选数据集上进行了预训练（参见补充方法“数据集”）。我们选择LSTM作为生成模型，因为其具有良好的泛化能力、灵活性以及捕捉中药特有局部特征的能力。虽然基于Transformer的模型如GPT在配体发现方面显示出潜力，但它们需要大量训练数据，并在生成过程中优先考虑全局分子拓扑结构 [12]，从而弱化了在我们方法中至关重要的药效团特征。在当前的预训练阶段，嵌入层和LSTM单元中的所有参数均通过梯度下降优化进行了优化（补充图S1B）[13]。最终的预训练模型在从ZINC15筛选数据集中拆分的测试集上表现稳定，实现了损失 <0.75，准确率 >0.75，以及困惑度 <2.5，表明有效

化学空间扩展能力评估。为了研究TCM-Generator是否能够扩展中药的化学空间，而不仅仅是“记忆”已有分子或在已知化学空间内进行插值，我们进行了外推生成测试[14]。我们首先通过计算每个分子的Morgan指纹表示来量化中药的化学空间（见补充方法“化学空间评估”）。然后对指纹数据应用主成分分析（PCA），提取前三个主成分（PC1、PC2和PC3），并将每个分子投影到由这些主轴形成的三维坐标系统中（补充图S2A-C）。为了可视化中药数据集的化学空间分布，我们应用了K-means聚类，将数据集划分为五个不同的簇（图2A，补充图S2 D）。接下来，我们排除了属于簇4的分子，该簇占分子总数的22.23%



manual experimental validation is time-consuming and labor-intensive, further slowing the progress of TCM-related drug discovery and limiting its clinical translation potential.

To address the above issues, we hypothesized that chemical language models, pre-trained with large chemical libraries and fine-tuned with TCM compound databases, can generate novel chemicals that exhibit biological, chemical, or pharmaceutical properties similar to those of original TCM molecules (TCM-like). We then developed TCM-navigator, the first deep learning-based, end-to-end workflow bridging between traditional TCM knowledge and modern drug discovery pipelines. TCM-navigator enables the generation, quality control, and physics-based evaluation of TCM-like compounds. It comprises three core modules: (i) TCM-generator, leverages the existing knowledge in the TCM knowledge base through transfer learning and an LSTM architecture, to expand and refine the original chemical space of the TCM database, allowing for flexible adaptation to target-specific TCM-like compound generation; (ii) TCM-identifier, the first quantitative TCM scoring model, leverages an Attentive FP framework with message passing neural networks (MPNNs), for compound quality assessment; and (iii) a binding affinity prediction module, integrating virtual screening and molecular dynamics (MD) simulations for structure-based evaluation. Beyond the enlarged and refined chemical space, the molecules in our TCM-like datasets exhibit improved synthesizability, drug-likeness, and favorable ADMET profiles compared to original TCM compounds, offering enhanced drug development potential.

To demonstrate the application of our TCM-like compounds, we prepared case study that comprises three important drug targets, including epidermal growth factor receptor (EGFR) for cancer, SARS-CoV-2 Main protease (Mpro) for viral infection, and Sirtuin 1 (SIRT1) for diabetes and metabolic diseases. These three targets are of important clinical applications for the treatment of various conditions, and have been targeted by TCM molecules [7–9]. In our case study, molecules from the TCM-like datasets show better predicted binding potential compared to TCM molecules known to interact with these targets, suggesting promising ligand candidates.

Beyond accelerating TCM-related drug development, more importantly, our workflow addresses a critical challenge in data-driven biology: standardizing and expanding limited datasets. Despite the proliferation of high-throughput technologies and deep learning-driven pipelines, small-to-medium-sized datasets in domains such as medicinal chemistry and synthetic biology frequently suffer from inadequate standardization, limited scalability, and incomplete data, rendering manual processing inefficient and error-prone. Our workflow tackles this by enabling *in silico* molecular generation, quality control, and physics-based evaluation, transforming sparse data into enriched, standardized resources for experimental exploration. This integrative approach unifies diverse biomedical domains, enabling scalable, efficient, and streamlined research.

## Results and Discussions

We proposed an end-to-end workflow — TCM-navigator — for TCM-like compound generation and evaluation, which integrates multiple data-driven and deep learning-based components. Our workflow is organized into three core design modules: two innovative components, including TCM-generator (Fig. 1A) and TCM-iidentifier (Fig. 1B), and physics-based evaluation (Fig. 1C). TCM-generator is a large-scale molecular generation module that employs a stagewise learning strategy consisting of three phases: pre-training, fine-tuning, and state-tuning. This phased approach empowers TCM-generator to progressively encode the chemical signatures of different compound libraries, achieving efficient feature extraction across stages [10]. TCM-Identifier is an AttentiveFP-based molecular distinguisher developed to evaluate the TCM-likeness of generated compounds [11]. As the first deep-learning model for TCM-like property assessment, TCM-identifier systematically captures global molecular information while preserving high interpretability. In the physics-based evaluation module, we employ two key criteria: (i) static binding affinity—molecular docking is employed to predict the binding conformations and evaluate the structural and chemical complementarity of drug-like molecules with target proteins; (ii) dynamic binding stability—MD simulations are conducted to assess the stability of shortlisted TCM-like compounds within the binding pocket. Our workflow establishes an innovative end-to-end paradigm for TCM-related drug development (Fig. 1).

## TCM-Generator: Chemical Language Model for generating TCM-like molecules

*Generation of the TCM-like molecules.* To enable the chemical language model to comprehensively learn the syntax and feature distribution of SMILES sequences, we pre-trained it on the ZINC15-filtered dataset (see Supplementary Methods ‘Datasets’). We selected LSTM as our generative model for its generalizability, flexibility, and ability to capture TCM-specific local features. While transformer-based models like GPTs show promise in ligand discovery, they require extensive training data and prioritize global molecular topology during the generation process [12], de-emphasizing pharmacophoric features central to our approach. All parameters in the embedding layer and LSTM cell were optimized via gradient descent optimization during the current pre-training phase (Supplementary Fig. S1B) [13]. The final pre-trained model demonstrated stable performance on the test set split from the ZINC15-filtered dataset, achieving a loss <0.75, accuracy >0.75, and perplexity <2.5, indicating effective learning of chemical representations. Subsequently, we utilized the pre-trained model as the point of initialization and fine-tuned it using the TCM dataset. The fine-tuning process selectively updated the fully connected layer parameters within the LSTM architecture. Model performance converged on the test set, achieving a loss <0.85, accuracy >0.68, and perplexity <0.25. Using the above fine-tuned model (TCM-Generator), we generated a large-scale TCM-like dataset containing 6 million molecular entries, expanding the number of entities in the TCM dataset by over 100-fold. Thereafter, we systematically characterized the dataset's chemical space and TCM-specific features to assess its utility in drug development.

*Evaluation of chemical space expansion capability.* To investigate whether TCM-Generator can expand the chemical space of TCM rather than merely ‘memorizing’ existing molecules or interpolating within the known chemical space, we conducted the extrapolative generation test [14]. We first quantified the chemical space of TCM by computing the Morgan Fingerprint representations for each molecule (see Supplementary Methods ‘Evaluation of Chemical Space’). Principal component analysis (PCA) was then applied to the fingerprint data to extract the top three principal components (PC1, PC2, and PC3), and each molecule was projected onto a three-dimensional coordinate system formed by these principal axes (Supplementary Fig. S2A- C). To visualize the chemical space distribution of the TCM dataset, we applied K-means clustering, dividing the dataset into five distinct clusters (Fig. 2A, Supplementary Fig. S2D). Next, we excluded molecules belonging to cluster 4, which comprises 22.23% of the molecules and is

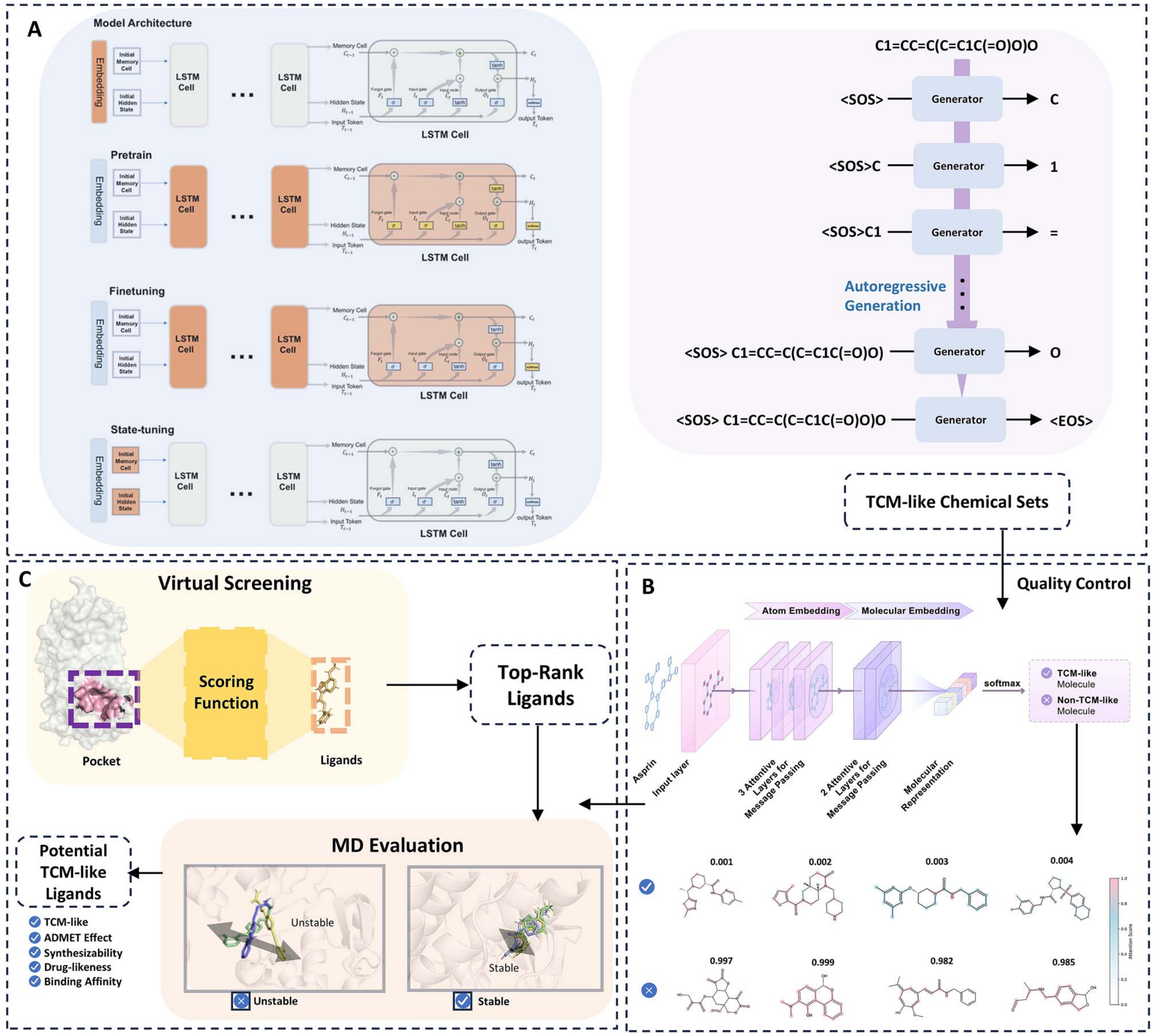


图1. 中医导航器工作流程概览。中医导航器通过多成分方法促进与中医相关的药物开发，提供从原创生成到质量控制及基于物理的评估的全面工作流程。该工作流程包括 (A) 中医生成器，(B) 中医识别器，以及 (C) 基于物理的评估器。我们基于LSTM架构开发了一种用于分子生成的迁移学习方法 (A)，设计了一种结合信息传递和注意力机制的模型来评估中医类特性 (B)，并整合分子对接与分子动力学模拟来验证配体结合 (C)。

位于由中药数据集定义的化学空间边缘（图2A），生成了一个名为中药子集的缩小数据集（图2B）。使用预训练模型作为初始化点，我们使用中药子集而不是完整的中药数据集对模型进行了200轮的微调。第200轮的参数被选为最终的生成模型参数，并使用该子集训练的模型生成了100,000个分子（图2C）。

有趣的是，当我们将生成的分子投影到 PC1、PC2 和 PC3 轴上时，我们观察到化学空间内的分子分布更加密集，占据同一空间的分子数量显著增加。值得注意的是，原先在 TCM 数据集中由簇 4 占据的区域，在 TCM 子集里被故意移除的部分，现在再次被高密度分子分布所填充（图 2C）。这些结果证实，该模型能够有效地在已知化学领域内放大分子密度，同时模拟同时扩展到化学空间的未知领域。

系统性探索未知化学空间的能力是全新设计的基石，它超越了现有知识库的限制，从而发现具有高临床潜力的新型治疗候选分子。我们证明了TCM-generator并不依赖记忆，而是通过外推测试学习底层模式。生成超出给定TCM数据集的新结构的能力，可能归因于对ZINC15筛选数据集的大规模预训练，这使模型接触到各种分子模体和表示，从而增强了其生成多样性。

对生成的类中药数据集的检查。对于生成的类中药数据集中的每个分子，我们测量了其与中药数据集中化合物的五个最高化学相似性分数，从而在随后的分析中实现前五相似性分数的更平滑、更具代表性的分布。



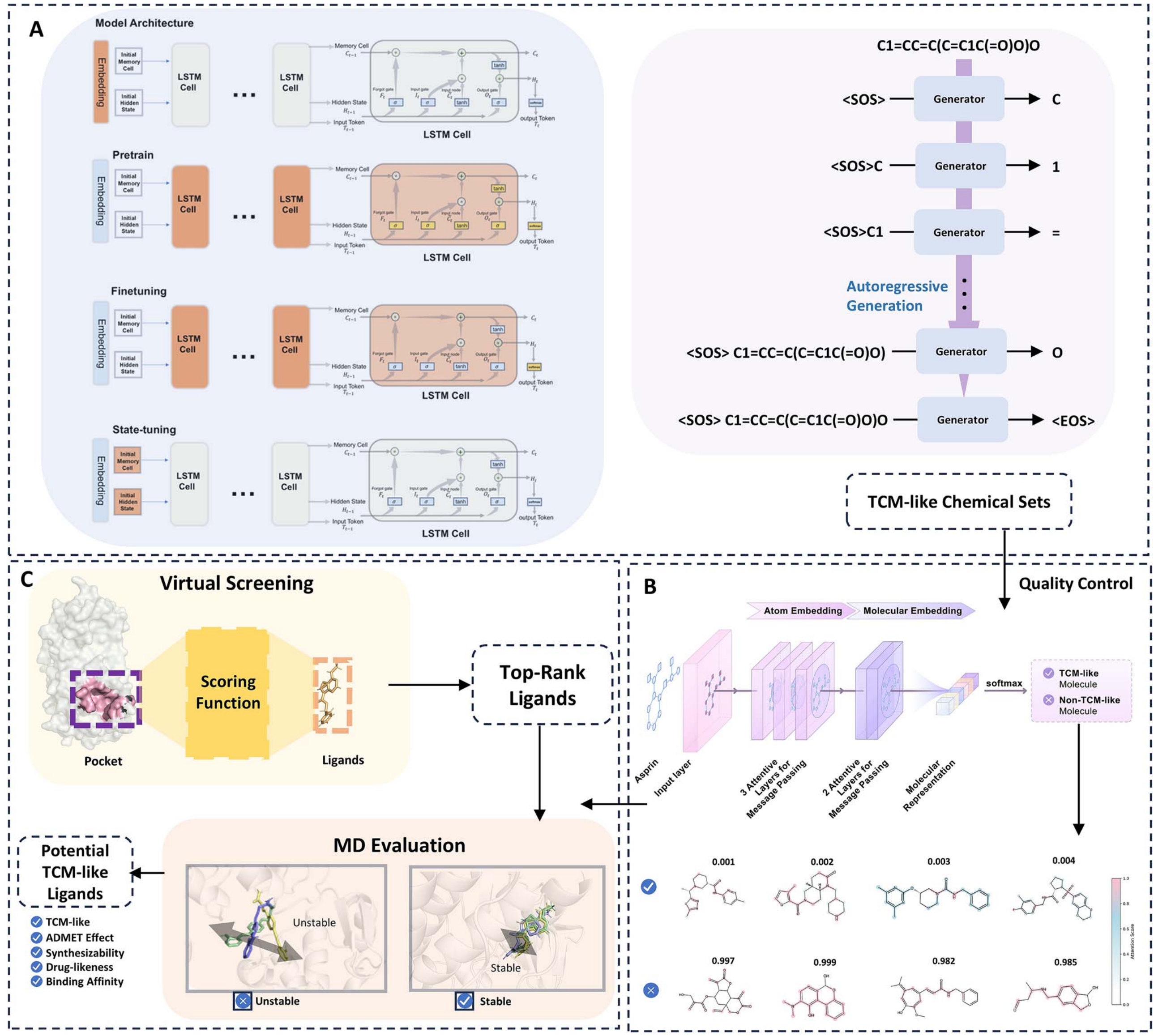

Figure 1. Overview of the TCM navigator workflow. TCM navigator facilitates TCM-related drug development through a multi-component approach, providing a comprehensive workflow from *de novo* generation to quality control and physics-based evaluation. This workflow consists of (A) TCM-generator, (B) TCM-identifier, and (C) the physics-based evaluator. We developed a transfer learning method based on the LSTM architecture for molecular generation (A), designed a model incorporating message passing and attention mechanisms to evaluate TCM-like properties (B), and integrated molecular docking with MD simulations to validate ligand binding (C).

located at the edge of the chemical space defined by the TCM dataset (Fig. 2A), generating a reduced dataset named as TCM Subset (Fig. 2B). Using the pre-trained model as the initialization point, we fine-tuned the model for 200 epochs using TCM Subset instead of the full TCM dataset. The parameters from epoch 200 were selected as the final generative model parameters, and 100 000 molecules were generated using the subset-trained model (Fig. 2C).

Interestingly, when we projected the generated molecules onto the PC1, PC2, and PC3 axes, we observed a denser molecular distribution within the chemical space, with significantly increased number of molecules occupying the same space. Notably, the region originally occupied by cluster 4 in the TCM dataset, which had been intentionally removed in the TCM Subset, was now repopulated with a high-density molecular distribution (Fig. 2C). These results confirm that the model can effectively amplify molecular density within known chemical domains while simultaneously expanding into unexplored regions of chemical space.

The capacity to systematically explore uncharted chemical space represents a cornerstone of *de novo* design, transcending the limitations of existing knowledge bases to unlock novel therapeutic candidates with high clinical potential. We demonstrated that TCM-generator does not rely on memorization but instead learns underlying patterns through the extrapolation test. The capability to generate novel structures beyond the given TCM dataset is likely attributed to the extensive pretraining on the ZINC15-filtered dataset, which exposed the model to a wide range of molecular motifs and representations, enhancing its generative diversity.

*Examination of the generated TCM-like dataset.* For each molecule in the generated TCM-like dataset, we measured its five highest chemical similarity scores to compounds in the TCM dataset, allowing for a smoother and more representative distribution of top similarity scores in the subsequent analysis. Such

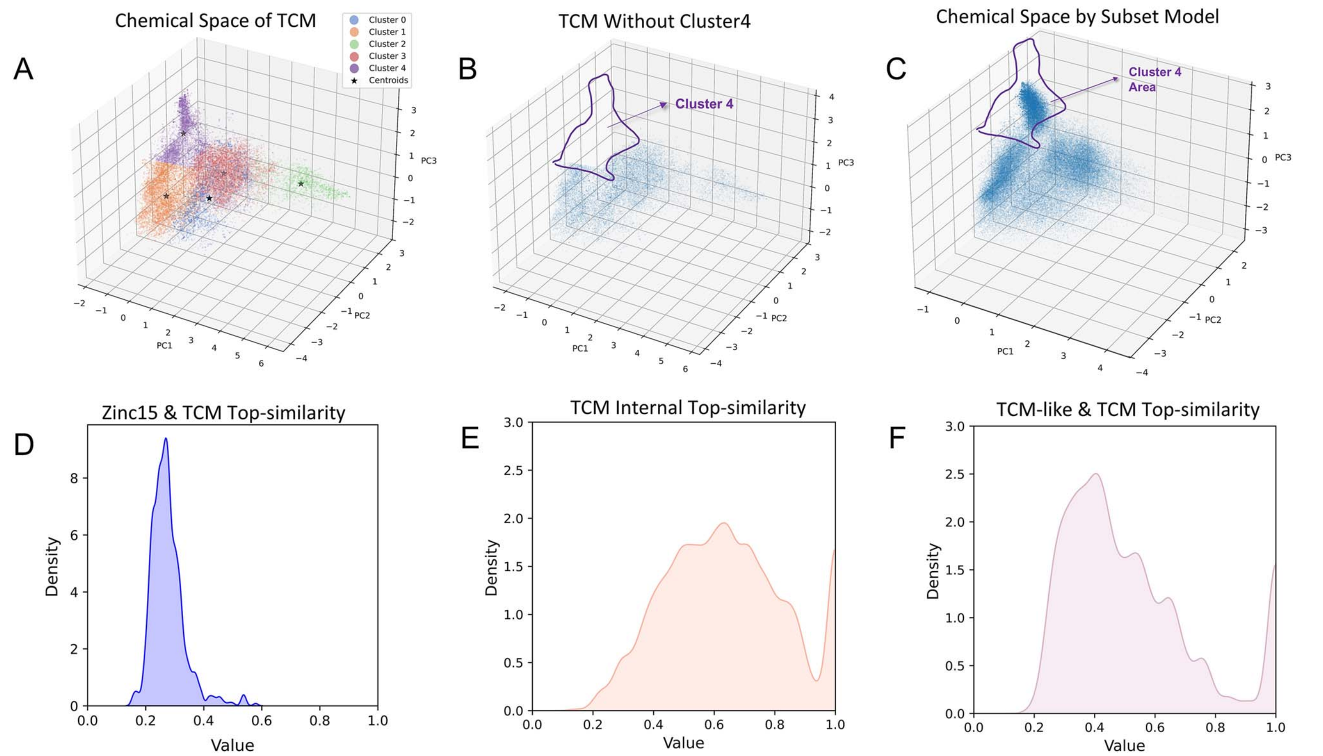


图 2. 对微调模型化学空间扩展及性能的评估。基于 Morgan 指纹的中药化学空间可视化 (A) 以及通过移除第 4 簇后的简化子集 (B)。子集模型生成的化学空间填补了第 4 簇的缺失区域 (C)。(D) 计算 ZINC15 与中药的前 5 相似性分布。(E) 中药内部前 5 相似性，从中药数据集中随机抽取的子集与剩余中药化合物计算得出。( F) 类中药化合物与中药之间的前 5 相似性。

top5-相似性分布也在ZINC15过滤后的数据集与TCM数据集之间（图2D），以及在TCM数据集内部（图2E）进行了计算，作为对照。我们的分析显示，TCM类与TCM数据集之间的top5-相似性分布（图2F）显著偏向高值，相比之下，ZINC15过滤后的数据集与TCM数据集之间的分布较低，并且类似于TCM数据集内部的相似性分布。这些结果表明，微调对模型生成行为具有强烈影响，生成的数据集（TCM类）可能捕捉到了微调集中（TCM）存在的特定领域特征。值得注意的是，一些TCM化合物是大型或复杂分子，其在小分子药物开发中的潜力有限。这样的分子也存在于TCM类数据集中。为了优化TCM类数据集的药物潜力和合成可行性，我们应用了筛选标准（a）（补充表S1），从而得到TCM类过滤后的数据集

为了获得统一的视角，我们对这些数据集中的所有分子进行了聚类，并识别出80个簇，每个簇代表一组具有相似ADMET相关特征的分子。80个簇的质心基于高维特征空间通过PCA投影到二维空间中（图3C）。此外，我们还计算了每个化学性质与前两个主成分（PC1和PC2）之间的相关性（补充图S3）。结果表明，PC1和PC2与Lipinski评分、分子量、logP、TPSA和溶解度的相关性最强，其次是QED、合成可及性（SA）以及与毒性相关的指标（hERG和LD50 zhu）。这些发现表明，这些性质是区分分子分布和簇结构的关键变量。将TCM-like数据集和TCM-like筛选数据集与TCM数据集进行比较，某些簇的比例分布发生了显著变化（图F）。

–

经过中药类筛选的数据集显示出改进的ADMET（吸收、分布、代谢、排泄和毒性）特性以及增强的类药物理化性质（图3A） [15, 16]（参见补充方法“ADMET与化学性质”）。值得注意的是，中药类筛选数据集在多个毒性相关预测指标上优于中药数据集，包括ClinTox、DILI、hERG和LD50 Zhu，这表明中药类化合物可能具有较低的毒性。在类药物特性方面，中药类筛选数据集中的化合物表现出更高分布的Lipinski评分以及更有利的QED特性（图3B）。此外，它们显示出较低的SA（合成可达性）评分，表明合成可行性有所提高。为了进一步探索中药、中药类及中药类筛选数据集之间的全球化学空间变化，从

此外，我们研究了那些比例显著增加或减少的簇，因为这种不成比例的变化表明化学空间的转移。我们提取了在中药样本过滤数据集中显示出不成比例高增加（图 3D）或减少（图 3E）的簇，以及每个簇的代表性分子，以进行进一步分析。我们发现，显著扩大的簇中的分子表现出独特的结构特征——例如，簇 43 富含共轭炔烃，簇 45 富含环氧基。这些扩大的簇也富含具有较低



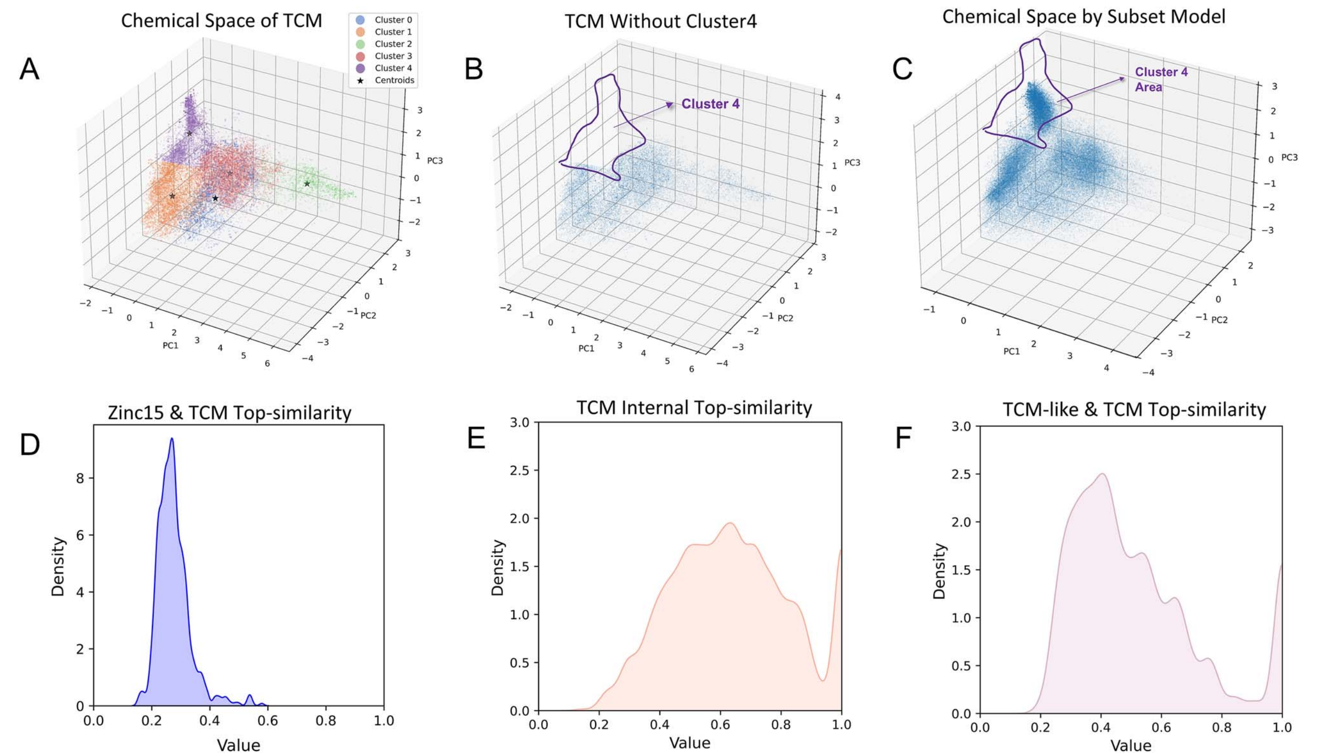


Figure 2. Evaluation of the fine-tuned Model's chemical space expansion and performance. Visualization of the chemical space of TCM (A) based on Morgan fingerprints and a reduced subset (B) by removing cluster 4. The chemical space generated by the subset model fills the missing region of cluster 4 (C). (D) Calculated Top5-similarity distributions between ZINC15 and TCM. (E) TCM internal Top5-similarity, a subset sampled from the TCM dataset calculated against the remaining TCM compounds. (F) Top5-similarity between TCM-like and TCM.

top5-similarity distributions were also calculated between the ZINC15-filtered and TCM datasets (Fig. 2D), as well as within the TCM dataset (Fig. 2E), as controls. Our analysis revealed that the top5-similarity distribution between the TCM-like and TCM datasets (Fig. 2F) is significantly skewed toward higher values than that between the ZINC15-filtered and TCM datasets, and resembles the intra-set similarity of the TCM dataset. These results suggest that fine-tuning had a strong influence on the model's generation behavior, and the generated dataset (TCM-like) likely captures domain-specific features present in the fine-tuning set (TCM). Notably, some TCM compounds are large or complex molecules that exhibit limited potential for small-molecule drug development. Such molecules are also present in the TCM-like dataset. To optimize the TCM-like dataset's drug potential and synthetic feasibility, we applied selection criteria (a) (Supplementary Table S1), resulting in the TCM-like-filtered dataset of 3.7 million entries.

The TCM-like-filtered dataset exhibits improved ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) profiles and enhanced drug-like physicochemical properties (Fig. 3A) [15, 16] (see Supplementary Methods 'ADMET and Chemical Properties'). Notably, the TCM-like-filtered dataset outperforms the TCM dataset in several toxicity-related prediction metrics, including ClinTox, DILI, hERG, and LD50_Zhu, suggesting that TCM-like compounds may show reduced toxicity. In terms of drug-like properties, the compounds in the TCM-like-filtered dataset exhibit a higher distribution of Lipinski scores and a more favorable QED profile (Fig. 3B). Additionally, they display lower SA (synthetic accessibility) scores, indicating improved synthetic feasibility. To further explore the global chemical space shifts across the TCM, TCM-like, and TCM-like filtered datasets from a unified perspective, we performed clustering on all molecules from these datasets, and identified 80 clusters with each cluster representing a group of molecules sharing similar ADMET-related features. The centroids of the 80 clusters were projected into a 2D space using PCA, based on the high-dimensional feature space (Fig. 3C). Additionally, we calculated the correlation between each chemical property and the first two principal components (PC1 and PC2) (Supplementary Fig. S3). The results showed that PC1 and PC2 were most strongly correlated with Lipinski score, molecular weight, logP, TPSA, and solubility, followed by QED, synthetic accessibility (SA), and toxicity-related metrics (hERG and LD50_zhu). These findings indicate that such properties are key variables in distinguishing molecular distributions and cluster structures. Comparing both the TCM-like dataset and the TCM-like filtered dataset to the TCM dataset, the proportional distribution of some clusters changed notably (Fig. 3C), while the number of molecules in most clusters increased significantly. These observations indicate that while the TCM-Generator substantially expands the number of molecules based on the TCM dataset, it also induces a noticeable shift in the chemical space.

Furthermore, we investigated the clusters whose proportions increased or decreased significantly, as such disproportionate changes suggest a shift in the chemical space. We extracted clusters that exhibited a disproportionately high increase (Fig. 3D) or decrease (Fig. 3E) in the TCM-like filtered dataset, along with representative molecules from each, for further analysis. We found that molecules in the significantly expanded clusters displayed distinct structural features—for example, Cluster 43 was enriched with conjugated alkynes, and Cluster 45 with epoxy groups. These expanded clusters were also enriched with molecules of lower

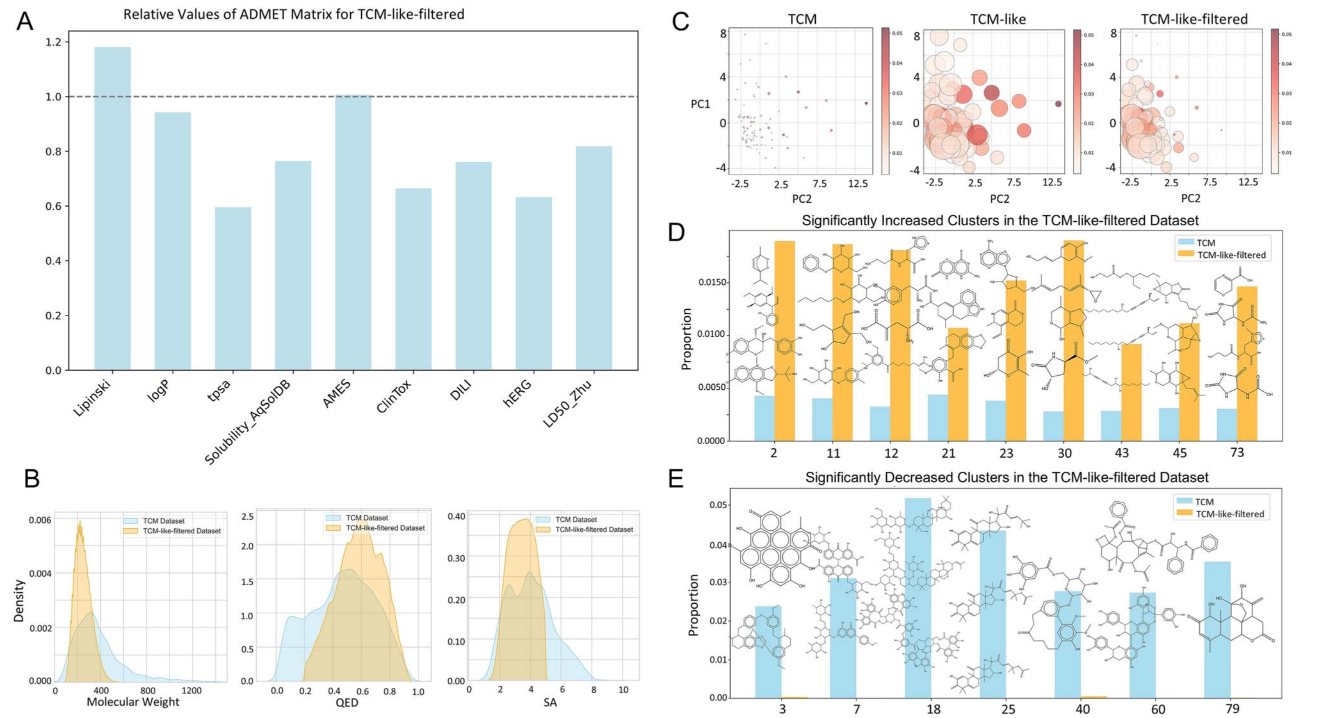


图 3. 类中药过滤数据集的系统评估。(A) 类中药过滤数据集的相对 ADMET 值。(B) 中药与类中药过滤数据集在分子量、QED 和 SA 分布上的比较。(C) 基于分子性质的中药、类中药和类中药过滤数据集的多中心聚类和 PCA 化学空间分析。分子被聚类为 80 个组，每个圆代表一个簇。颜色深浅（由浅到深红色）表示每个簇中中药分子的比例，而圆的大小表示每个簇中三个数据集分子相对比例。(D) 与 (E) 类中药过滤数据集中相对于中药数据集显著增加的簇和显著减少的簇，以及每个簇的代表性分子示例。

分子量和更高的药物相似性（图3D）。相反，减少的簇主要由分子量较高且环系更复杂的分子组成（图3E）。这些发现提供了簇级证据，表明TCM-Generator不仅扩展了化学空间，还优化了化学空间，这也反过来解释了在TCM样式筛选数据集中观察到的合成可及性提高、毒性降低和药物相似性增强。

## TCM标识符：中医相似性测量

保留中药特性对于中药类化合物的生成是必不可少的。然而，这些特性复杂且抽象，涵盖多种理化和生物学特征[17]。为了有效量化中药相似性，需要一个标准化的端到端模型。为了增强我们的中药识别器的泛化能力，并有效捕获全局和几何分子特征，我们采用了AttentiveFP架构 [11]，，该架构将注意力机制与消息传递神经网络（MPNN）以及迭代的原子和分子级表示结合（图1B）。我们从ZINC15筛选数据集中随机采样分子作为负样本，并使用中药数据集中的分子作为正样本以构建完整数据集。然后，我们将10%的数据划分为测试集。由于类别不平衡——负样本约为正样本的五倍——我们在训练过程中应用了加权学习率策略。在充分训练之后

可视化提供了对模型决策过程的可解释性。

我们首先使用TCM-identifier计算了ZINC15筛选数据集（抽样）、TCM数据集、从COCONUT [18],获得的天然产物数据集以及TCM-like数据集中的每个分子的TCM-like得分。然后，我们拟合并比较了这些数据集的TCM-like得分分布（图4A–D）。ZINC15筛选数据集中超过95%的分子TCM-like得分在0–0.05范围内，而TCM数据集中超过90%的分子得分在0.8到1之间。天然产物数据集呈现双峰分布，峰值分别位于0–0.05（>16%）和0.95–1（>48%），反映其由TCM/TCM-like和不喜欢TCM的化合物混合组成。TCM-like数据集与TCM数据集非常相似，有超过90%的分子得分在0.8到1之间，确认其强烈的TCM-like特性。这些分布突出了TCM-Identifier捕捉TCM独有但并非所有天然产物普遍存在的结构特征的能力。

最近，基于 Transformer 的模型如 ChemBERTa [19, 20] 和 MolGPT [21] 已经被证明能够成功生成新分子。为了与我们的 TCM-Generator 进行比较，使用与 TCM-Generator 相似的参数和相同的训练数据训练了一个单层 Transformer 解码器模型。我们选择使用内部开发的 Transformer 模型，而不是现有的预训练选项，因为它允许我们比较具有可比参数的模型并



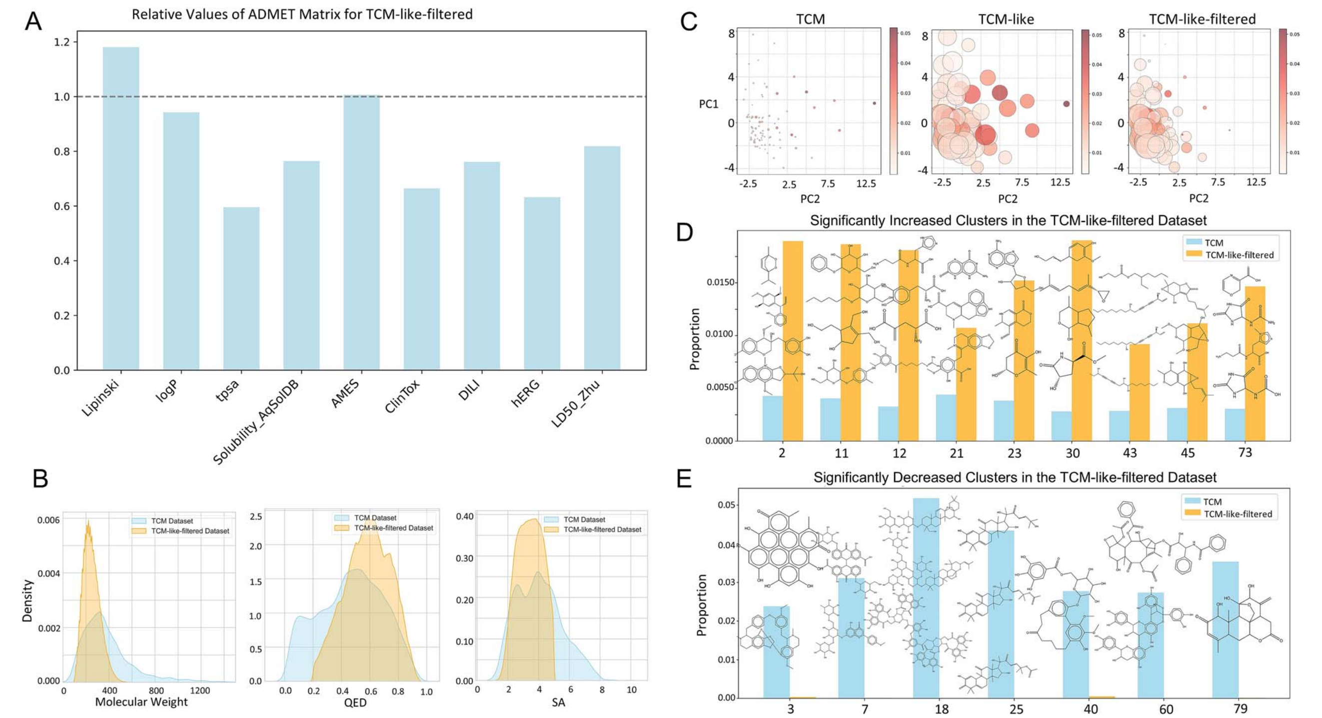


Figure 3. Systematic evaluation of the TCM-like-filtered dataset. (A) Relative ADMET values of the TCM-like-filtered dataset. (B) Distribution of molecular weight, QED, and SA between the TCM and the TCM-like-filtered. (C) Chemical space analysis using multi-Center clustering and PCA for TCM, TCM-like, and TCM-like-filtered datasets based on molecular properties. Molecules were clustered into 80 groups, with each circle representing a cluster. The color intensity (from light to deep red) represents the proportion of TCM molecules in each cluster, while the circle size represents the relative proportions of molecules from three datasets in each cluster. (D) Significantly increased clusters and (E) significantly decreased clusters in the TCM-like-filtered dataset compared to the TCM dataset, along with representative molecular examples from each cluster.

molecular weight and higher drug-likeness (Fig. 3D). In contrast, the reduced clusters primarily consisted of molecules with higher molecular weights and more complex ring systems (Fig. 3E). These findings provide cluster-level evidence that the TCM-Generator leads to both an expansion and refinement of the chemical space, which also in turn explains the improved synthetic accessibility, reduced toxicity, and enhanced drug-likeness observed in the TCM-like filtered dataset.

## TCM-identifier: Measurement of TCM-likeness

Retaining TCM properties is essential for TCM-like compound generation. However, these properties are complex and abstract, encompassing multiple physicochemical and biological features [17]. To efficiently quantify TCM-likeness, a standardized end-to-end model is required. To enhance the generalization ability of our TCM-Identifier and effectively capture global and geometric molecular features, we adopted AttentiveFP architecture [11], which integrates attention mechanisms with message-passing neural networks (MPNN) and iterative atom- and molecular-level representations (Fig. 1B). We randomly sampled molecules from the ZINC15-filtered dataset as negative samples and used molecules from the TCM dataset as positive samples to construct the full dataset. We then split 10% of the data as the test set. Due to the class imbalance—negative samples being approximately five times more than positive samples—we applied a weighted learning rate strategy during training. After sufficient training, the TCM-identifier achieved an AUPR (area under the precision-recall curve) of 0.978 on the test set for the task of identifying TCM-like molecules (Supplementary Fig. S4). Additionally, attention score visualization provides interpretability into the model's decision-making process.

We first used the TCM-identifier to calculate the TCM-like score for each molecule in the ZINC15-filtered dataset (sampled), the TCM dataset, the natural product dataset obtained from COCONUT [18], and the TCM-like dataset. We then fitted and compared the TCM-like score distributions of these datasets (Fig. 4A–D). The ZINC15-filtered dataset had over 95% of molecules with TCM-like scores in the 0–0.05 range, whereas more than 90% of molecules in the TCM dataset scored between 0.8 and 1. The natural product dataset displayed a bimodal distribution, with peaks at 0–0.05 (>16%) and 0.95–1 (>48%), reflecting its mixed composition of both TCM/TCM-like and TCM-dislike compounds. The TCM-like dataset closely resembled the TCM dataset, with over 90% of molecules scoring between 0.8 and 1, confirming its strong TCM-like characteristics. These distributions underscore the capability of TCM-Identifier to capture structural features that are unique in TCM but not universally present in all natural products. We hypothesize that, since TCM compounds are primarily used for therapeutic purposes, TCM and TCM-like molecules inherently possess safety- and efficacy-related characteristics that are often lacking in non-medicinal natural products.

Recently, transformer-based models like ChemBERTa [19, 20] and MolGPT [21] have demonstrated to be able to successfully generate novel molecules. For comparison with our TCM-Generator, a single-layer transformer-decoder model was trained using similar parameters to those in TCM-Generator with the same training data. An in-house transformer model was employed in preference to available pre-trained options, as it allows us to compare models with comparable parameters and

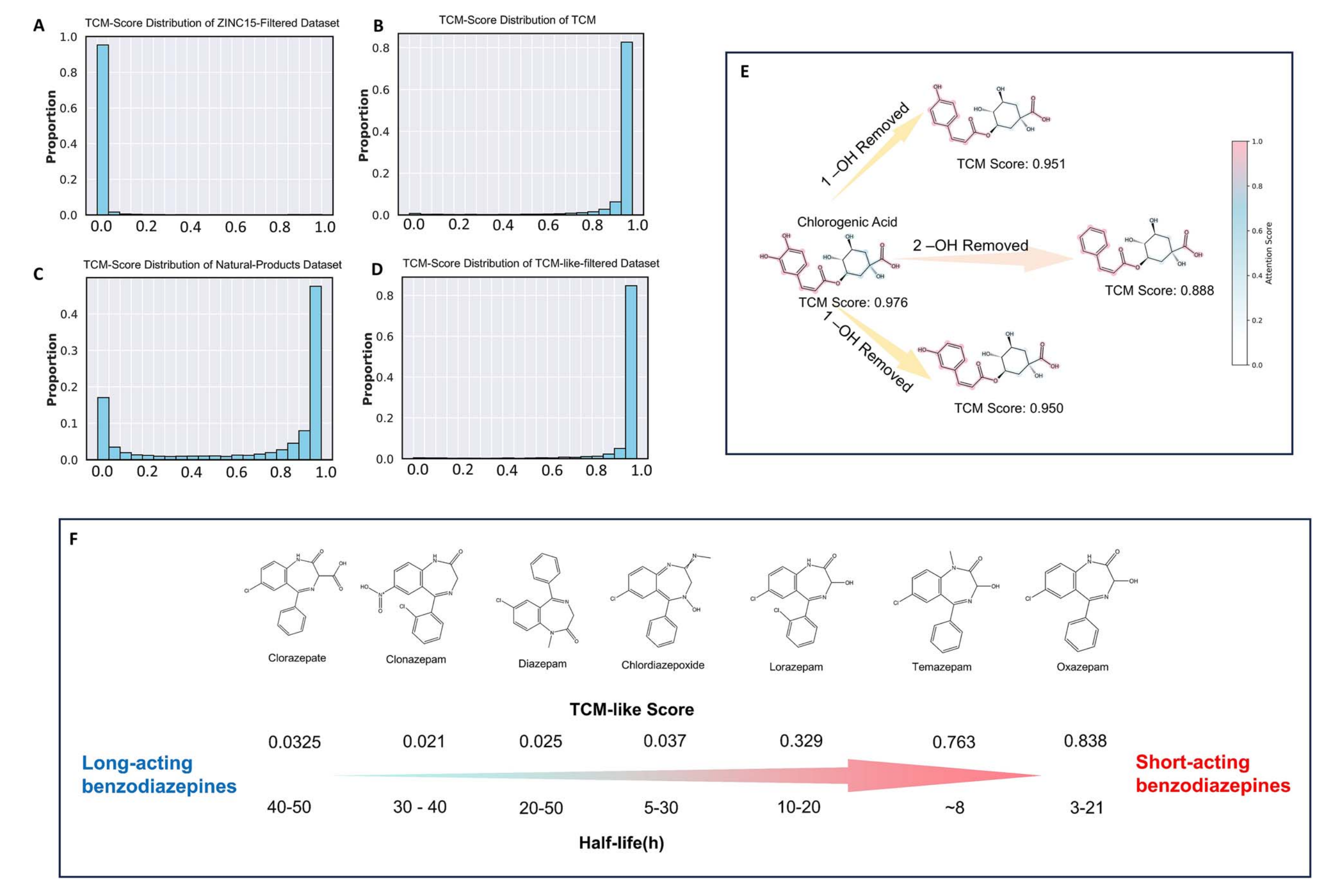


图 4. 使用中药（TCM）识别器的中药样评分测量。(A) ZINC15筛选数据集、(B) 中药数据集、(C) 天然产物数据集以及 (D) 中药样筛选数据集中的中药样评分分布。(E) 中药识别器的注意力主要集中在酚、酯键和羧基功能团特征上。去除酚羟基会导致分子酸性降低、功能改变以及中药样评分下降。(F) 常见长效、中效和短效苯二氮卓类药物及其半衰期的中药样评分。

训练。在应用选择标准(a)（补充表S1）后，使用基于Transformer的模型生成了389,574种独特的化合物以进行比较。计算了基于Transformer模型生成的化合物的ADMET性质和类中药评分，并与TCM-Generator筛选的类中药数据集进行比较（补充图S5）。基于Transformer模型生成的数据集显示出的ADMET性质与TCM-Generator生成的化合物相似（补充图S5A）。类中药评分的分布也与TCM-Generator化合物相当，其中超过90%的化合物的类中药评分高于0.8（补充图S5B）。有趣的是，由TCM-Generator生成的类中药化合物中，类中药评分超过0.95的比例略高于基于Transformer模型生成的化合物。这表明TCM-Generator对类中药性质具有更细致的理解，但这也可能是

我们进一步使用一个代表性的中药分子——绿原酸（中药相似度评分：0.976），一种酚酸类中药成分[22]，对TCM-Identifier的可解释性进行了研究。酚酸类是中药中常见的成分，其酸性来源于酚羟基和羧基。我们的TCM-Identifier对邻二酚和羧基赋予了更高的注意力评分

与环己烷核心相比的基团——关键的酸性片段（图4E），显示了其识别分类关键化学基团的能力。此外，该模型对微小的结构变化表现出敏感性，这可以通过去除酚羟基后TCM类得分的下降得到证明。

有趣的是，我们在另一个案例研究中观察到中医（TCM）类评分与药物半衰期之间的潜在关联，这支持了我们最初的假设，即与中医相关的特性可能与其他各种理化和生物学特性相关。具体而言，我们通过苯二氮䓬类药物——一种广泛使用的合成药物类别——来说明这种联系。值得注意的是，尽管苯二氮䓬类药物在结构上相似，但其中医类评分存在差异，并且这些评分与其半衰期呈正相关，而半衰期是药物排泄的一个关键指标。例如，Temazepam 和 Oxazepam 的中医类评分较高（分别为 0.763 和 0.838），其半衰期分别显著较短（分别为 ~8 和 3–21 小时），相比之下，Clorazepate 的中医类评分较低（0.033），且半衰期更长（40–50 小时）（图 4F）。有趣的是，尽管 Oxazepam 和 Clorazepate 的结构仅略有差异 —— 在酰胺邻近位置分别有一个羟基和羧基——这一微小变化导致



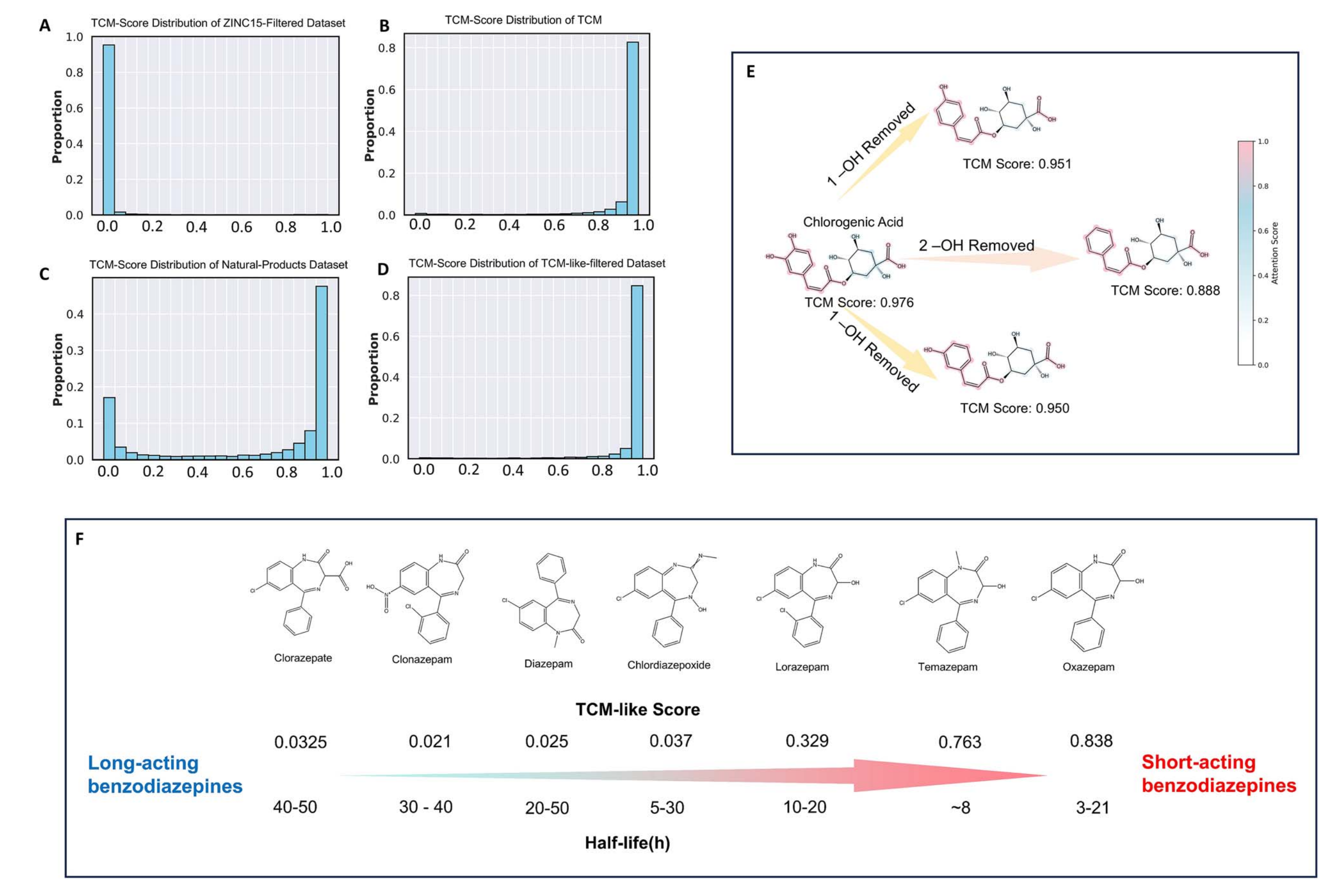

Figure 4. TCM-like score measurement with TCM identifier. Distribution of TCM-like scores in the (A) ZINC15-filtered dataset, (B) TCM dataset, (C) natural products dataset, and (D) TCM-like-filtered dataset. (E) TCM Identifier's attention primarily focuses on the phenol, ester bonds, and carboxyl functional group features. Removal of the phenolic hydroxyl group results in a decrease in the molecule's acidity, functional changes, and a decrease in the TCM-like score. (F) TCM-like scores of common long-acting, medium-acting, and short-acting benzodiazepine drugs and their half-life.

training. After applying selection criteria (a) (Supplementary Table S1), 389,574 unique compounds were generated with the transformer-based model for comparison. The ADMET properties and TCM-like scores were computed for compounds generated by the transformer-based model and compared to the TCM-like filtered dataset from TCM-Generator (Supplementary Fig. S5). The dataset generated by the transformer-based model exhibits ADMET properties similar to those of the compounds generated by TCM-Generator (Supplementary Fig. S5A). The distribution of TCM-like scores was also comparable to that of the TCM-Generator compounds, with more than 90% of the compounds having TCM-like scores above 0.8 (Supplementary Fig. S5B). Interestingly, TCM-like compounds generated by TCM-Generator have a slightly higher proportion of scores above 0.95 than those generated by the transformer-based model. This suggests that TCM-Generator has a more nuanced understanding of TCM-like properties, but this may also be due to differences in the number of compounds generated. Overall, a single-layer transformer-based model does not offer any significant advantage over the TCM-Generator for the compounds generated, although a more complex model may offer better performance.

We further investigated the interpretability of the TCM-Identifier using a representative TCM molecule—chlorogenic acid (TCM-like score: 0.976), a phenolic acid-type TCM ingredient [22]. Phenolic acids, which are common in TCM, derive their acidity from phenol hydroxyl and carboxyl groups. Our TCM-Identifier assigned higher attention scores to the catechol and carboxyl groups—key acidic fragments—compared to the cyclohexane core (Fig. 4E), demonstrating its ability to identify critical chemical moieties for classification. Moreover, the model showed sensitivity to subtle structural variations, as evidenced by the decrease in TCM-like scores when the phenol hydroxyl groups were removed.

Interestingly, we observed a potential link between TCM-like scores and drug half-lives in another case study, supporting our initial hypothesis that TCM-related properties may correlate with various other physicochemical and biological characteristics. Specifically, we illustrate this connection using benzodiazepines—a widely used class of synthetic drugs [23]. Notably, despite their structural similarities, benzodiazepines exhibit distinct TCM-like scores that positively correlate with their half-lives, a key indicator of drug excretion. For example, Temazepam and Oxazepam, which have higher TCM-like scores (0.763 and 0.838), exhibit significantly shorter half-lives (~8 and 3–21 hours, respectively), compared to Clorazepate, which has a lower TCM-like score (0.033) and a much longer half-life (40–50 hours) (Fig. 4F). Interestingly, although Oxazepam and Clorazepate differ only slightly in structure—by a hydroxyl and a carboxyl group adjacent to the amide—this minor variation results in notable differences in both their TCM-like scores and pharmacokinetics. This case highlights the TCM-like score's sensitivity to subtle structural features and its potential predictive power for key pharmacokinetic properties. We propose that this correlation arises from the fact that TCM-like scores are grounded in natural compound characteristics, which are

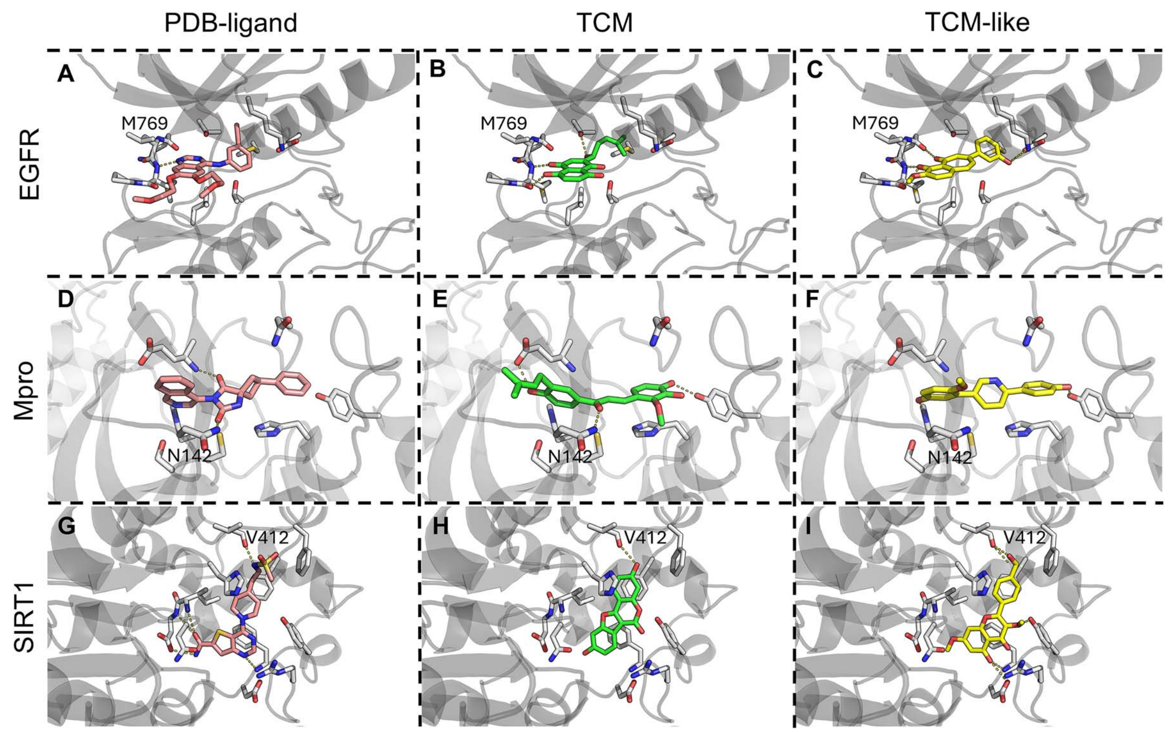


图5. EGFR、Mpro 和 SIRT1 的 PDB-配体、TCM 和类 TCM 化合物的选定对接构象。(A) EGFR 的 PDB 配体 Erlotinib，(B) EGFR TCM Shikonin 的对接构象，(C) EGFR-TCML1。(D) Mpro 的 PDB 配体，(E) Mpro TCM 的对接构象，(F) Mpro-TCML1。(G) SIRT1 的 PDB 配体，(H) SIRT1 TCM 的对接构象，(I) SIRT1-TCML1。

通常与更快的新陈代谢和排泄相关，使得有利的中医类评分表明半衰期较短。

## 生成特定靶点的中医类数据集

选择了三个靶点：EGFR、Mpro 和 SIRT1，以展示我们模型生成特定靶点中类似中药（TCM）数据集的能力。选择这些靶点是基于它们的临床相关性、具有已报道结合口袋的晶体结构可用于对接和分子动力学模拟，以及存在已验证的中药分子用于状态调整。我们分别收集了针对 EGFR 、 Mpro 和 SIRT1 的 20 、 22 和 24 个实验验证的中药配体。这些靶点的中药配体数量相对较少，使它们成为扩展其靶点特异性类似中药分子化学空间的理想候选。为了解决配体数据有限的问题，我们保留了 TCM-Generator 的核心框架，并采用状态调节策略，在训练期间仅更新初始隐藏状态和记忆单元（见补充方法“TCM-Generator”，补充图 S1）。为了确保生成的化合物具有良好的类药性潜力并与靶点具有结合亲和力，我们应用了

这三个数据集的类中药评分呈现单峰分布，其中超过75%的分子得分在0.8到1之间，显示出类中药特性（补充图 S4B–E，补充表 S1）。这些分布与中药数据集相比略有偏差，这是由于应用了选择标准 (b) 所致。这些结果

证明状态微调能够有效保留嵌入在类似中药复合物分布中的知识，在ADMET/理化性质谱图上产生可忽略的差异（补充图S6）。在保留类似中药复合物特征的ADMET和理化性质的同时，状态微调过程有效地将TCM-Generator的生成分布与特定靶点中药数据集的SMILES序列对齐。具体而言，经过状态微调后，TCM-Generator的下一步标记预测准确性相比仅在通用中药数据集上训练的模型显著提高，在不同靶点上分别从0.742提高到0.817、0.748提高到0.800、0.690提高到0.735。TCM-Generator的多阶段训练策略能够以低计算成本实现高效的状态微调，并且可以普遍适用于任何已有中药配体报道的靶蛋白，同时始终保持ADMET和理化性质分布的稳定特征

## 基于物理的中药类化合物结合亲和力评估

针对EGFR、Mpro和SIRT的目标特异性中药（TCM）和类似中药的数据集被对接到各自的PDB结构，对于每种蛋白质，评估了前500个类似中药化合物的对接评分。对于这三种蛋白质，前500个类似中药化合物显示出与中药相似或更好的对接评分（补充图S7）。这提示TCM生成器具备生成可能与中药相当或更优结合能力的化合物的潜力。对于每种蛋白质，选择了一个中药和一个类似中药化合物，以及晶体配体（图5），进行进一步分析和分子动力学模拟。

对所选化合物对接结果的分析。EGFR参考配体厄洛替尼（图5A）结合到EGFR激酶结构域的ATP口袋，与骨架形成氢键



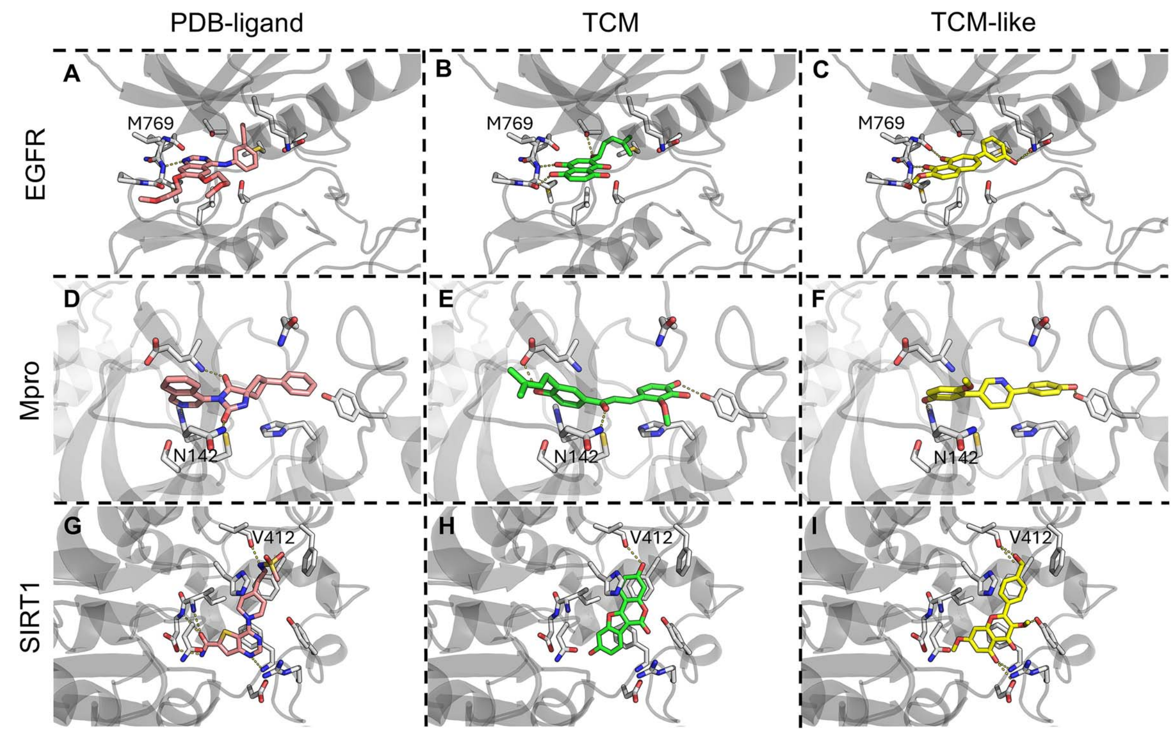


Figure 5. Selected docking poses of PDB-ligand, TCM, and TCM-like compounds for EGFR, Mpro and SIRT1. (A) PDB ligand of EGFR, erlotinib, (B) docking pose of EGFR TCM, Shikonin, and (C) EGFR-TCML1. (D) PDB ligand of Mpro, (E) docking pose of Mpro TCM, and (F) Mpro-TCML1. (G) PDB ligand of SIRT1, (H) docking pose of SIRT1 TCM, and (I) SIRT1-TCML1.

generally associated with faster metabolism and excretion, making favorable TCM-like scores indicative of shorter half-lives.

## Generation of target-specific TCM-like datasets

Three targets, EGFR, Mpro, and SIRT1 were selected to demonstrate our model's ability to generate target-specific TCM-like datasets. They were chosen based on their clinical relevance, the availability of crystal structures with reported binding pocket for docking and MD simulations, and the presence of verified TCM molecules for state-turning. We collected 20, 22, and 24 experimentally validated TCM ligands for EGFR, Mpro, and SIRT1, respectively. The relatively small number of TCM ligands for these targets made them ideal candidates for expanding their target-specific TCM-like molecules chemical space. To address the issue of limited ligand data, we preserved TCM-Generator's core framework and employed a state-tuning strategy, by only updating the initial hidden states and memory cells during training (see Supplementary Methods 'TCM-Generator', Supplementary Fig. S1). To ensure the generated compounds possess favorable drug-like potential and binding affinity to the target, we applied selection criteria (b) (Supplementary Table S1), yielding 6179, 4528, and 6179 molecules for the EGFR, Mpro, and SIRT1, respectively. All datasets exhibited favorable ADMET/physicochemical profiles, with higher drug potential and lower synthetic complexity compared to the TCM dataset (Supplementary Fig. S6).

The TCM-like scores of these three datasets displayed a unimodal distribution, with over 75% of molecules scoring between 0.8 and 1, demonstrating TCM-like characteristics (Supplementary Fig. S4B–E., Supplementary Table S1). These distributions exhibited slight deviations from the TCM dataset, due to the application of selection criteria (b). These results demonstrate that state-tuning effectively preserves the knowledge embedded in the TCM-like compound distribution, yielding negligible differences in ADMET/physicochemical profiles (Supplementary Fig. S6). While preserving the ADMET and physicochemical profiles characteristic of TCM-like compounds, the state-tuning process effectively aligned the generation distribution of the TCM-Generator with the SMILES sequences of target-specific TCM datasets. Specifically, after state-tuning, the next-token prediction accuracy of the TCM-generator improved significantly compared to the model trained solely on the general TCM dataset, increasing from 0.742 to 0.817, 0.748 to 0.800, and 0.690 to 0.735 across different targets. TCM-Generator's multi-stage training strategy enables efficient state- tuning with low computational cost and can be universally adapted to any target protein with reported TCM ligands, while consistently preserving ADMET and physicochemical property distributions that remain largely unaffected by variations in the training set composition.

## Physics-based binding affinity evaluation of TCM-like compounds

The target-specific TCM and TCM-like datasets for EGFR, Mpro, and SIRT were docked to their respective PDB structures, and the docking scores for the top 500 TCM-like compounds for each protein were evaluated. For all three proteins, the top 500 TCM-like compounds showed similar or improved docking scores compared to their TCM (Supplementary Fig. S7). This hints at the capability of the TCM-Generator to generate compounds that can potentially bind comparable or better than TCM. For each protein, one TCM and one TCM-like compound, alongside the crystal ligands (Fig. 5), were selected for further analysis and MD simulations.

*Analysis of docking results for selected compounds*. The EGFR reference ligand, Erlotinib (Fig. 5A), binds to the ATP pocket of EGFR kinase domain, forming a hydrogen bond with the backbone of

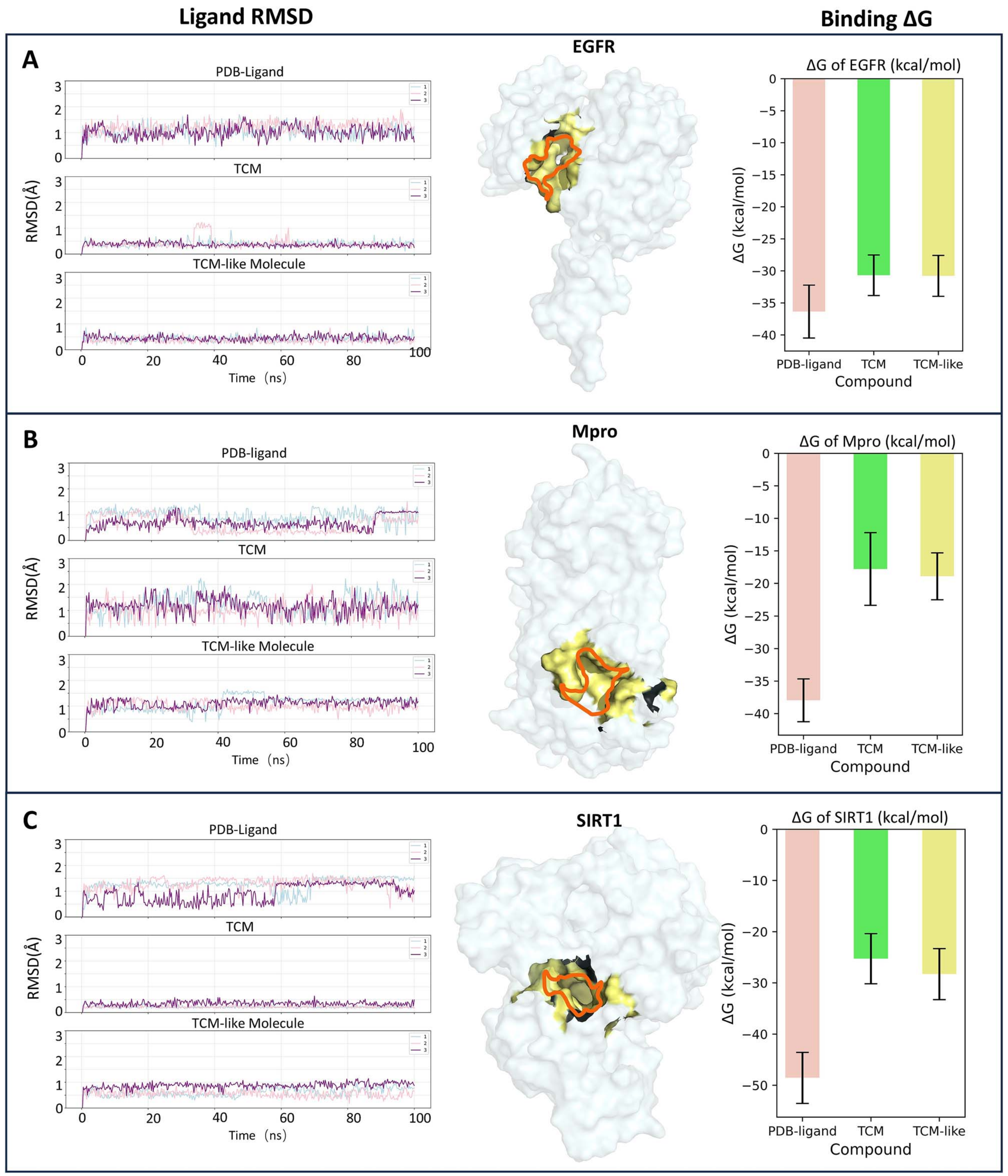


图6. 基于分子动力学(MD)的配体结合亲和力评估。对于EGFR (A)、Mpro (B)和SIRT1 (C)，左侧显示了PDB配体、中药分子及类中药分子的RMSD值，曲线表示三次独立重复实验。中间面板显示了每个蛋白的表面表示，并高亮了目标结合口袋。右侧显示了每个目标所选化合物的MMGBSA结合自由能。

M769。所选的中药（图5B）与EGFR的对接得分为–7.68 kcal/mol，而中药类化合物（图5C）的对接得分为 –9.21 kcal/mol。用于EGFR的所选中药和中药类化合物均为杂环化合物，可与M769的主链形成氢键，类似于PDB配体。对于Mpro，所选的中药和中药类化合物（图5E, F）的对接得分为 –6.37 kcal/mol 分别为 – 7.84 kcal/mol。尽管所选的类中药化合物不像 PDB 配体和所选中药化合物那样与 N142 形成氢键，但其对接评分并未受影响，这可能是因为其可旋转键数量较少，为三个，而所选中药则有六个可旋转键。对于 SIRT1，所选的中药和类中药化合物（图 5H, I）的对接评分为 –8.33 kcal/mol



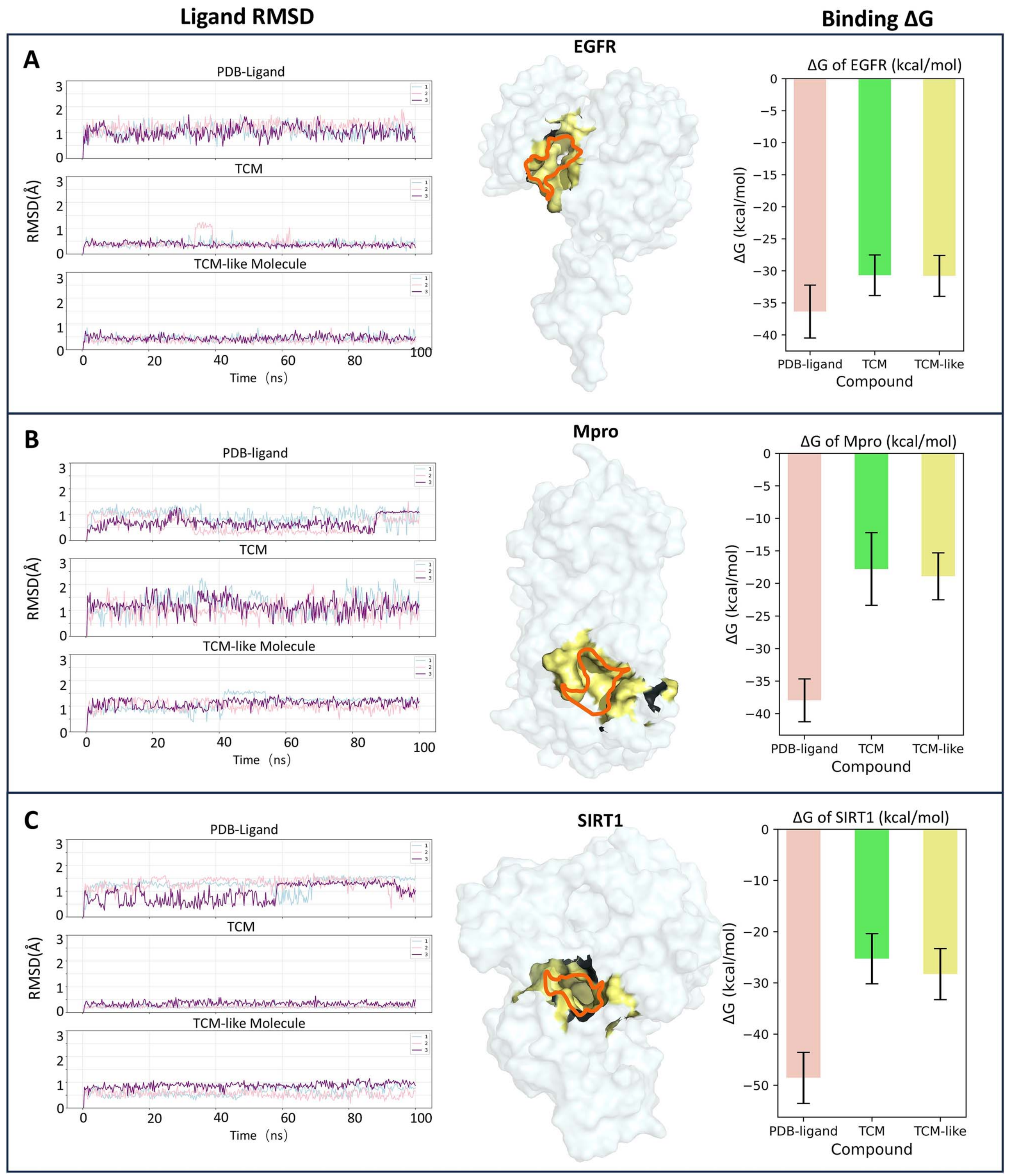


Figure 6. MD-based evaluation of ligand binding affinity. For EGFR (A), Mpro (B), and SIRT1 (C), the RMSD values of the PDB ligand, TCM molecule, and TCM-like molecule are shown on the left, with the lines representing three independent replicates. The middle panel shows the surface representation of each protein, with the targeted binding pocket highlighted. On the right are the MMGBSA binding free energies of the selected compounds for each target.

M769. The selected TCM (Fig. 5B) docks to EGFR with a docking score of –7.68 kcal/mol, while the TCM-like compound (Fig. 5C) has a docking score of –9.21 kcal/mol. The selected TCM and TCM-like compounds for EGFR are both heterocyclic compounds that form hydrogen bonds with the backbone of M769, similar to the PDB ligand. For Mpro, the selected TCM and TCM-like compounds (Fig. 5E, F) show docking scores of –6.37 kcal/mol and – 7.84 kcal/mol, respectively. Although the selected TCM-like compound does not form a hydrogen bond with N142 as the PDB ligand and the selected TCM compound do, its docking score is not compromised, probably due to its lower number of rotatable bonds of three compared to the selected TCM with six rotatable bonds. For SIRT1, the selected TCM and TCM-like compounds (Fig. 5H, I) show docking scores of –8.33 kcal/mol

分别为 – 8.67 kcal/mol。SIRT1的这三种选定化合物都与V412的主链形成氢键。这些类中药化合物的例子说明了所生成化合物与相应PDB配体和中药化合物在对接构象上的相似性。

MD模拟分析。EGFR的PDB配体和厄洛替尼的MD模拟显示出较高的RMSD值（图6A，平均RMSD：1.09 Å），相比之下，选定的中药（TCM）和类中药化合物的平均RMSD分别为0.37 Å和0.40 Å。然而，正如从PDB配体预期的那样，厄洛替尼具有最佳的平均MMGBSA $\Delta G$，为 –36.35 kcal/mol。类中药化合物显示出与中药化合物相似的平均结合$\Delta G$，为 –30.78 kcal/mol，而中药化合物为–30.68 kcal/mol。此外，对M769主链的氢键分析显示，我们的类中药化合物在67%的帧中形成稳定的相互作用，而中药分子为46%，厄洛替尼为17%。对于Mpro（图6B），PDB配体、中药和类中药化合物的平均RMSD分别为0.74 Å、1.12 Å和1.07 Å，平均MMGBSA结合 $\Delta G$分别为 –37.95 kcal/mol、 –17.80 kcal/mol和– 18.91 kcal/mol。与EGFR类似，SIRT1的PDB配体也显示出较高的平均值

## 结论

在这项研究中，我们建立了TCM-Navigator，这是一个以数据驱动和深度学习为基础的中药相关药物开发工作流程，包括TCM-Generator、TCM-Identifier和基于物理的评估器。据我们所知，这是首个生成类中药数据集并定量定义中药相似性的工作。TCM-Generator采用分阶段训练策略，包括预训练、微调和状态调优，从而实现对有限中药知识的高效利用。生成的类中药数据集满足现代药物设计的要求，并在ADMET性质上优于中药。TCM-Identifier利用MPNN和自注意力机制，从原子级细节到整体特征准确捕捉分子特征，为涉及类中药分子的研究提供了可靠的度量标准和宝贵的指导。最后，我们的基于物理的方法通过对接和分子动力学模拟评估生成化合物的计算机体内结合能力。

我们的方法将中医与现代药物发现相结合，结合了历史上的生物活性和现代理化标准。

## 方法

方法的详细描述可以在补充方法中找到，其中以下内容有更详细的描述：‘数据集’、‘TCM-Generator’、‘使用TCM-Generator生成化合物’、‘化学空间评估’、‘ADMET和化学性质’、‘TCM-Identifier’、‘基于Transformer的模型’、‘分子对接’和‘分子动力学模拟’。

**关键点**

- 我们推出了TCM-navigator，这是一种基于深度学习的端到端工作流程，用于生成和评估类中药化合物，以用于药物开发。
- TCM-navigator 由 TCM-generator、TCM-identifier 以及基于物理的评估器组成。
- TCM-Generator 是一个用于生成类似中药分子的 LSTM 模型。
- TCM-Identifier 是第一个用于中药样分子鉴定的基于深度学习的模型。
- 使用该工作流程，我们生成了蛋白质特异性的中药类似化合物，并且对所选中药类似化合物的计算机模拟评价显示其亲和力优于或可与中药化合物相媲美。

## 致谢

作者感谢生物信息学研究所（A*STAR）的资金支持。

## 作者贡献

陈飞英和范浩构思并设计了该项目。陈飞英和林俊宇生成并分析了结果，所有作者（陈飞英、林俊宇、李明宇和范浩）撰写了手稿。

## 补充数据

补充数据可在《生物信息学简报》在线获取。

利益冲突：无申报。

## 资金

未申报。

## 数据可用性

TCM-Generator 和 TCM-Identifier 可执行文件可在 GitHub 上公开下载，https://github.com/cfy2yue/TCM-Navigator。生成的类似中药复合物 SMILES 可在 https://zenodo.org/uploads/15518561 获取。

## 参考文献

1. Cheung FTCM. 中国制造。Nature 2011;480:S82–3. https://doi.org/10.1038/480S82a 2. Lv Q, Chen G, He H. 等. TCMBank：通过智能文本挖掘连接最大的草药、化学成分、靶蛋白及相关疾病。Chem Sci 2023;14:10684–701. https://doi.org/10.1039/D3SC02139D 3. Tu Y. 青蒿素的发现及中医药的馈赠。Nat Med 2011;17:1217–20. https://doi.org/10.1038/nm.2471 4. Xu H-Y, Zhang Y-Q, Liu Z-M. 等. ETCM：传统中医药百科全书。Nucleic Acids Res 2019;47:D976–82. https://doi.org/10.1093/nar/gky987



and – 8.67 kcal/mol, respectively. All three selected compounds of SIRT1 form a hydrogen bond with the backbone of V412. These examples of TCM-like compounds illustrate the comparable docking conformation of the generated compounds compared to the corresponding PDB ligands and TCM compounds.

*Analysis of MD simulations.* The MD simulations of EGFR PDB ligand, Erlotinib, show higher RMSD values (Fig. 6A, average RMSD: 1.09 Å) compared to the selected TCM and TCM-like compounds (average RMSD: 0.37 Å and 0.40 Å, respectively). However, Erlotinib has the best average MMGBSA $\Delta G$ of −36.35 kcal/mol, as expected from the PDB ligand. The TCM-like compound exhibits a similar average binding $\Delta G$ of −30.78 kcal/mol to that of the TCM compound (−30.68 kcal/mol). Furthermore, analysis of hydrogen bonding with the backbone of M769 revealed that our TCM-like compound forms a stable interaction in 67% of the frames, compared to 46% for the TCM molecule and 17% for Erlotinib. For Mpro (Fig. 6B), the average RMSD of the PDB ligands, TCM, and TCM-like compounds are 0.74 Å, 1.12 Å, and 1.07 Å, respectively, and the average MMGBSA binding $\Delta G$ are −37.95 kcal/mol, −17.80 kcal/-mol, and – 18.91 kcal/mol, respectively. Similar to EGFR, the SIRT1 PDB ligand also shows a higher average RMSD of 1.16 Å compared to TCM (0.26 Å) and TCM-like compound (0.68 Å) (Fig. 6C). Nevertheless, the average MMGBSA binding $\Delta G$ of the PDB ligand is the most favourable (−48.57 kcal/mol), outperforming the TCM (−25.27 kcal/mol) and the TCM-like compound (−28.29 kcal/mol). In all three case studies, the TCM-like compounds demonstrate performance comparable to their corresponding TCM compounds though not surpassing the PDB ligands, which are typically highly optimized for binding affinity. This highlights the ability of our methods to generate TCM-like compounds with *in-silico* performance on par with TCM compounds.

## Conclusions

In this study, we established TCM-Navigator, a data-driven and deep learning-based workflow for TCM-related drug development consisting of TCM-Generator, TCM-Identifier, and a physics-based evaluator. To the best of our knowledge, this is the first work to generate TCM-like datasets and quantitatively define TCM-likeness. The TCM-Generator used a stagewise training strategy consisting of pre-training, fine-tuning, and state-tuning, enabling the efficient utilization of the limited TCM knowledge. The generated TCM-like dataset meets modern drug design requirements with improved ADMET properties over TCM. The TCM-Identifier leveraged MPNN and self-attention mechanisms to accurately capture molecular features from atomic-level details to global characteristics, providing a reliable metric and valuable guidance for studies involving TCM-like molecules. Lastly, our physics-based approach employs docking and MD simulations to evaluate the *in-silico* binding of generated compounds.

Our approach bridges TCM with contemporary drug discovery, combining historical bioactivity with modern physicochemical standards.

## Methods

The detailed description of the methods can be found in Supplementary Methods where the following were described in greater details: ‘Datasets’, ‘TCM-Generator’, ‘Compounds generation with TCM-Generator’, ‘Evaluation of Chemical Space’, ‘ADMET and Chemical Properties’, ‘TCM-Identifier’, ‘Transformer-based Model’, ‘Molecular Docking’ and ‘MD simulations’.

**Key Points**

- We introduced TCM-navigator, a deep learning-based, end-to-end workflow for the generation and evaluation of TCM-like compounds for drug development.
- TCM-navigator consists of TCM-generator, TCM-identifier, and a physics-based evaluator.
- TCM-Generator is an LSTM model for generating TCM-like molecules.
- TCM-Identifier is the first deep-learning-based model for TCM-like molecule identification.
- Using the workflow, we generated protein-specific TCM-like compounds and *in*-silico evaluation of selected TCM-like compounds yields better or comparable affinity with respect to TCM compounds.

## Acknowledgements

The authors acknowledge funding support from Bioinformatics Institute, A*STAR.

## Author contributions

Feiying Chen and Hao Fan conceptualised and designed the project. Feiying Chen and Victor Jun Yu Lim generated and analyzed results, and all authors (Feiying Chen, Victor Jun Yu Lim, Mingyu Li, and Hao Fan) wrote the manuscript.

## Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

Conflict of interest: None declared.

## Funding

None declared.

## Data availability

The TCM-Generator and TCM-Identifier executables are publicly available for download at GitHub, https://github.com/cfy2yue/TCM-Navigator. The generated TCM-like compounds SMILES are available at https://zenodo.org/uploads/15518561.

## References

1. Cheung FTCM. Made in China. *Nature* 2011;**480**:S82–3. https://doi.org/10.1038/480S82a
2. Lv Q, Chen G, He H. *et al.* TCMBank: Bridges between the largest herbal medicines, chemical ingredients, target proteins, and associated diseases with intelligence text mining. *Chem Sci* 2023;**14**:10684–701. https://doi.org/10.1039/D3SC02139D
3. Tu Y. The discovery of artemisinin (qinghaosu) and gifts from Chinese medicine. *Nat Med* 2011;**17**:1217–20. https://doi.org/10.1038/nm.2471
4. Xu H-Y, Zhang Y-Q, Liu Z-M. *et al.* ETCM: An encyclopaedia of traditional Chinese medicine. *Nucleic Acids Res* 2019;**47**:D976–82. https://doi.org/10.1093/nar/gky987

5. Chen X, Zhou H, Liu YB 等. 传统中药数据库及其在机制研究和处方验证中的应用. 英国药理学杂志 2006;149:1092–103. https://doi.org/10.1038/sj.bjp.0706945 6. Li M, Zhang J. 聚焦于利用大数据和人工智能：从传统中药资源革新药物发现. 化学科学 2023;14:10628–30. https://doi.org/10.1039/D3SC90185H 7. Han L, Wang B, Sun K 等. 一种用于探索传统中药药效物质的 SARS-CoV-2 M pro 荧光传感器. 分析师 2024;149:3585–95. https://doi.org/10.1039/D4AN00372A 8. Wu Z, Zhu Q, Zhang Y 等. EGFR 相关通路在传统中药 (TCM)-1 诱导的前列腺癌细胞生长抑制、自噬和凋亡中的作用. 分子医学报告 (Spandidos 出版社), 2018. https://doi.org/10.3892/mmr.2018.8818 9. Luttens A, Gullberg H, Abdurakhmanov E 等. 超大虚拟筛选 id

14. Hatakeyama-Sato K, Oyaizu K. 材料信息学中用于外推预测的生成模型。ACS Omega 2021;6:14566–74. https://doi.org/10.1021/acsomega.1c01716 15. Tetko IV, Bruneau P, Mewes H-W. 等. 我们能估计ADME–毒性预测的准确性吗？Drug Discov Today 2006;11:700–7. https://doi.org/10.1016/j.drudis.2006.06.013 16. Swanson K, Walther P, Leitz J. 等. ADMET-AI：用于评估大规模化学库的机器学习ADMET平台。Bioinformatics 2024;40:btae416. https://doi.org/10.1093/bioinformatics/btae416 17. Schoretsanitis G, Deligiannidis KM, Paulzen M. 等. 精神药物与口服避孕药之间的药物-药物相互作用。Expert Opin Drug Metab Toxicol 2022;18:395–411. https://doi.org/10.1080/17425255.2022.2106214 18. Sorokina M, Merseburger P, Rajan K. 等. COCONUT 在线：开放天然产物数据库集合。J Chem 2021;13:2. https://doi.org/10.1186/s13321-020-00478-9 19. Chithrananda S, Grand G





5. Chen X, Zhou H, Liu YB. *et al.* Database of traditional Chinese medicine and its application to studies of mechanism and to prescription validation. *Br J Pharmacol* 2006;**149**:1092–103. https://doi.org/10.1038/sj.bjp.0706945
6. Li M, Zhang J. A focus on harnessing big data and artificial intelligence: Revolutionizing drug discovery from traditional Chinese medicine sources. *Chem Sci* 2023;**14**:10628–30. https://doi.org/10.1039/D3SC90185H
7. Han L, Wang B, Sun K. *et al.* A SARS-CoV-2 $M^{pro}$ fluorescent sensor for exploring pharmacodynamic substances from traditional Chinese medicine. *Analyst* 2024;**149**:3585–95. https://doi.org/10.1039/D4AN00372A
8. Wu Z, Zhu Q, Zhang Y. *et al. EGFR-Associated Pathways Involved in Traditional Chinese Medicine (TCM)-1-Induced Cell Growth Inhibition, Autophagy and Apoptosis in Prostate Cancer*. Rep: Mol. Med (Spandidos Publications), 2018. https://doi.org/10.3892/mmr.2018.8818.
9. Luttens A, Gullberg H, Abdurakhmanov E. *et al.* Ultralarge virtual screening identifies SARS-CoV-2 main protease inhibitors with broad-Spectrum activity against coronaviruses. *J Am Chem Soc* 2022;**144**:2905–20. https://doi.org/10.1021/jacs.1c08402
10. Amabilino S, Pogány P, Pickett SD. *et al.* Guidelines for recurrent neural network transfer learning-based molecular generation of focused libraries. *J Chem Inf Model* 2020;**60**:5699–713. https://doi.org/10.1021/acs.jcim.0c00343
11. Xiong Z, Wang D, Liu X. *et al.* Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism. *J Med Chem* 2020;**63**:8749–60. https://doi.org/10.1021/acs.jmedchem.9b00959
12. Özçelik R, De Ruiter S, Criscuolo E. *et al.* Chemical language modeling with structured state space sequence models. *Nat Commun* 2024;**15**:6176. https://doi.org/10.1038/s41467-024-50469-9
13. Tay DWP, Yeo NZX, Adaikkappan K. *et al.* 67 million natural product-like compound database generated via molecular language processing. *Sci Data* 2023;**10**:296.
14. Hatakeyama-Sato K, Oyaizu K. Generative models for extrapolation prediction in materials informatics. *ACS Omega* 2021;**6**:14566–74. https://doi.org/10.1021/acsomega.1c01716
15. Tetko IV, Bruneau P, Mewes H-W. *et al.* Can we estimate the accuracy of ADME–tox predictions? *Drug Discov Today* 2006;**11**:700–7. https://doi.org/10.1016/j.drudis.2006.06.013
16. Swanson K, Walther P, Leitz J. *et al.* ADMET-AI: A machine learning ADMET platform for evaluation of large-scale chemical libraries. *Bioinformatics* 2024;**40**:btae416. https://doi.org/10.1093/bioinformatics/btae416
17. Schoretsanitis G, Deligiannidis KM, Paulzen M. *et al.* Drug-drug interactions between psychotropic medications and oral contraceptives. *Expert Opin Drug Metab Toxicol* 2022;**18**:395–411. https://doi.org/10.1080/17425255.2022.2106214
18. Sorokina M, Merseburger P, Rajan K. *et al.* COCONUT online: Collection of open natural products database. *J Chem* 2021;**13**:2. https://doi.org/10.1186/s13321-020-00478-9
19. Chithrananda S, Grand G, Ramsundar B. ChemBERTa: *Large-Scale Self-Supervised Pretraining for Molecular Property Prediction*. ArXiv Preprint 2020.
20. Ahmad W, Simon E, Chithrananda S. *et al. ChemBERTa-2: Towards Chemical Foundation* Models, ArXiv Preprint 2022.
21. Bagal V, Aggarwal R, Vinod PK. *et al.* MolGPT: Molecular generation using a transformer-decoder model. *J Chem Inf Model* 2022;**62**:2064–76. https://doi.org/10.1021/acs.jcim.1c00600
22. Naveed M, Hejazi V, Abbas M. *et al.* Chlorogenic acid (CGA): A pharmacological review and call for further research. *Biomed Pharmacother Biomedecine Pharmacother* 2018;**97**:67–74. https://doi.org/10.1016/j.biopha.2017.10.064
23. Greenblatt DJ, Shader RI, Divoll M. *et al.* Benzodiazepines: A summary of pharmacokinetic properties. *Br J Clin Pharmacol* 1981;**11 Suppl 1**:11S–6. https://doi.org/10.1111/j.1365-2125.1981.tb01833.x


*Briefings in Bioinformatics*, 2025, **26(5)**, bbaf498
https://doi.org/10.1093/bib/bbaf498
Problem Solving Protocol